Optimized Image Combination Selection and Confidence-Guided DSM Fusion for 3-D Reconstruction From Multiview Satellite Imagery

Shuting Yang, Hao Chen, Fachuan He, and Wen Chen

***Abstract*—The 3-D reconstruction based on multiview satellite imagery effectively leverages the redundant information to mitigate the effects of occlusions, noise, and other factors. Deep learning-based multiview stereo (MVS) methods are primarily applied to tri-stereo satellite data. Traditional MVS methods often fail to fully consider the comprehensiveness of image coverage and the information gain during image pair selection. Additionally, digital surface model (DSM) fusion typically relies on simple median filtering (MF) or its variants, which lack the flexibility required for handling heterogeneous information. To address these challenges, this article presents a novel MVS method based on optimized image combination selection and confidence-guided DSM fusion. First, to ensure comprehensive coverage and low redundancy, multiview images are partitioned based on their meta-parameters, and the interval boundaries are adjusted to ensure that the images are evenly distributed across the intervals. Next, one image is randomly selected from each interval for combination. A hierarchical heuristic strategy is then designed to select the optimized image combination that satisfies conditions such as intersection angles and sun angles. Subsequently, a stereo matching method based on iterative optimization of hierarchical graph structure consistency (GSC) cost is employed to generate multiple DSMs. Meanwhile, confidence features and estimated confidence are enhanced using contextual information. Finally, a confidence-guided DSM fusion method is designed, which effectively suppresses the interference of outliers by balancing local statistical properties with single-point confidence. Experimental results demonstrate that the proposed method outperforms mainstream methods, achieving an average mean absolute error (MAE) of 1.31 m, a root-mean-square error (RMSE) of 2.51 m, a median height error (MHE) of 0.43 m, and a COMP of 70.27%.**

***Index Terms*—3-D reconstruction, digital surface model (DSM) fusion, image combination selection, multiview satellite imagery.**

I. Introduction

As THE number of high-resolution optical satellites with stereo mapping capabilities increases, significant advancements have been made in 3-D reconstruction of the Earth's surface, making it a research hotspot in the fields of photogrammetry and computer vision. This progress holds considerable importance in various geospatial applications, including urban planning, navigation, and environmental monitoring [1], [2], [3], [4], [5]. Compared to aerial imagery, satellite imagery offers unique advantages in terms of data acquisition cost and coverage range. It enables large-scale reconstruction with consistent spatial resolution. Additionally, the satellite's orbital movement allows for the capture of images from different perspectives, providing rich spatial redundancy for 3-D reconstruction. This redundancy effectively mitigates the challenges posed by occlusions, shadows, and atmospheric noise, thus improving the accuracy of digital surface models (DSMs) [6], [7].

There are two different ways of capturing multiview stereo (MVS) images: one involves using multiple line-scan cameras mounted on a satellite to capture images simultaneously [8], while the other captures multiple images of the same scene within a defined time window using a single line-scan camera [9]. Given the data acquisition cost and the benefit of utilizing images from more perspectives, the second method is typically employed in experiments.

Existing satellite MVS methods primarily include commercial software packages [10], [11], [12], deep learning-based methods [13], [14], [15], [16], and traditional satellite stereo reconstruction pipelines [9], [17], [18]. The high cost and limited flexibility of commercial software packages, coupled with the inability of deep learning methods to fully meet the demands of satellite MVS tasks and their substantial requirements for training samples and GPU memory, make traditional satellite stereo reconstruction pipelines still an effective solution, particularly when handling larger or more numerous satellite images.

The classical MVS reconstruction method based on satellite imagery follows a pair-wise approach: 1) multiple views are grouped into pairs; 2) each pair is processed using a two-view stereo method to generate a DSM or point cloud; and 3) the pair-wise reconstruction results are fused to produce the final output. The quality of the final reconstruction is determined by the quality of the pair-wise DSMs, which depends on several factors, including the intersection angle between images, sun angle differences, time differences in image acqui-

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sition, and environmental conditions. In the traditional MVS reconstruction pipeline, in addition to the stereo matching step, two crucial steps ensure high-quality reconstruction: 1) the optimization of image pair selection and 2) the DSM fusion method.

In the image pair selection step, various factors influence the quality of the image pairs. This complex task is typically addressed through heuristic methods that take image meta-parameters into account. In [9], a heuristic approach is used to rank all possible image pairs, ensuring that the first pairs of the list yield results with higher completeness measure. In [17], an image and rational polynomial coefficient (RPC) simulation tool is employed to consider all possible geometric configurations and precompute a map that encodes the reconstruction quality in relation to the meta-parameters of the image pairs. This map serves as a proxy for the quality of the real image pairs. The heuristic approach of ranking image pairs or evaluating their quality using simulation tools to select the best-performing pairs for subsequent processing may lead to issues of incomplete coverage and redundant information. Specifically, multiple well-performing image pairs could correspond to nearly identical viewing angles, providing no additional information while increasing computational costs. Therefore, during image pair selection, it is crucial to consider the comprehensiveness of the included viewing angles based on imaging geometry while minimizing computational overhead and ensuring reconstruction accuracy.

At the other end of the pair-wise MVS pipeline, various methods can be applied to merge or aggregate DSMs [19], [20]. Typically, simple techniques such as weighted averaging (WA), median filtering (MF), or their variants are used for DSM fusion. However, these methods can be affected by poor-quality DSMs, and WA relies on precise weight information, which may introduce biases. Therefore, it is essential to design DSM fusion algorithms that effectively combine the complementary strengths of multiple DSMs, mitigating the impact of outliers and improving the accuracy of 3-D reconstruction in urban areas.

In general, traditional satellite MVS methods primarily focus on evaluating the quality of individual stereo pairs during the image pair selection stage, aiming to prioritize those with potentially higher reconstruction accuracy. However, such approaches often neglect the redundancy between stereo pairs and the completeness of spatial coverage. In multiview 3-D reconstruction tasks, relying solely on a limited number of top-ranked pairs can lead to imbalanced viewing angle distributions and concentrated occlusions, ultimately resulting in local reconstruction failures [9], [17], [21], [22], [23]. Moreover, during the DSM fusion stage, traditional satellite MVS methods typically fail to leverage the complementary information across multiple DSMs. Simple statistical strategies, such as MF and its variants, are commonly adopted, which are prone to degradation caused by low-quality reconstructions. In complex urban scenes characterized by dense occlusions and significant elevation variations, these fusion approaches tend to yield inaccurate decisions, thereby compromising the structural completeness and accuracy of the reconstructed models [17], [19], [20], [24]. To address these limitations, this article presents a novel satellite MVS method based on optimized image combination selection and confidence-guided DSM fusion. First, satellite images are divided into multiple subintervals according to imaging geometry, with the interval boundaries adjusted to ensure an even distribution of multiview images across the intervals. One image is then randomly selected from each interval to form a combination, and a hierarchical heuristic strategy is designed based on image meta-parameters to select the optimized image combination. Next, a 3-D reconstruction method based on iterative optimization of hierarchical graph structure consistency cost (IOHGSCM) is employed to generate multiple DSMs. Confidence is constructed by integrating auxiliary information generated during the MVS reconstruction process, and both confidence features and estimated confidence are enhanced by incorporating contextual information. Finally, the height values are ranked and clustered based on the confidence results. By balancing local statistical properties with single-point confidence, the interference of outliers is suppressed, leading to effective DSM fusion. Experimental results validate the effectiveness of the proposed hierarchical heuristic multiview image selection strategy and the confidence-guided DSM fusion method. Moreover, the proposed satellite MVS method outperforms mainstream methods. The main contributions of this article are summarized as follows.

1) An optimized image combination selection method based on rule-driven partitioning and hierarchical heuristic strategies is proposed to alleviate reconstruction instability caused by redundant stereo pair selection and insufficient spatial coverage.
2) A confidence-guided DSM fusion method is proposed to address the limitations of traditional DSM fusion approaches in distinguishing DSM quality and their sensitivity to occlusions and noise.

The remainder of this article is organized as follows. Section II presents a brief review of related work on 3-D reconstruction from optical satellite image pairs and multiview satellite images. The details of our proposed method are described in Section III. Section IV presents extensive experiments conducted on multiple datasets to demonstrate the effectiveness of the proposed method. Finally, Section V provides a conclusion.

II. Related Work

This section provides a review of two closely related topics: 3-D reconstruction from optical satellite image pairs and 3-D reconstruction from multiview optical satellite images.

A. 3-D Reconstruction From Optical Satellite Image Pairs

Differing from common aerial imagery, most optical satellite images are captured using line-array scanners with a push-broom approach. The RPC model is a widely used geometric model in the high-resolution satellite domain, with negligible accuracy loss [25]. The advantages of the RPC model have greatly benefited research in 3-D reconstruction from satellite images.

Most methods for 3-D reconstruction based on optical satellite image pairs first perform epipolar rectification and then apply stereo matching to obtain dense conjugate points

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[26]. The epipolar geometry is based on the RPC model. Hirschmuller [27] proposed the widely used semiglobal matching (SGM) algorithm, which provides a good balance between matching quality and computational efficiency, forming the core of many satellite image 3-D reconstruction solutions. The CATENA utilizes SGM with distributed optimization to automatically generate high-resolution DSMs [28]. Additionally, many commercial software packages use SGM-based algorithms for 3-D information extraction. Yang et al. [29] proposed a stereo matching method based on iterative optimization of hierarchical graph structure consistency (GSC) cost, which overcomes radiometric differences by constructing and mapping the graph structure, while optimizing boundary disparities using the visibility term and the disparity discontinuity term. Tatar et al. [30] introduced an object-based stereo matching method and content-guided disparity refinement for high-resolution satellite image 3-D reconstruction, addressing boundary blurring issues. Lee et al. [31] proposed a robust dense stereo matching algorithm using adaptive random walk with restart, which is resilient to illumination changes and performs well in occluded and disparity discontinuity regions. Zhao et al. [32] presented a double propagation stereo matching method for urban 3-D reconstruction from satellite images, demonstrating robustness to disparity discontinuities, inclined surfaces, and occlusions. There are also a few methods that rectify stereo images by approximating the push-broom geometry of small cropped image tiles using a pinhole model and then performing standard stereo matching. The satellite stereo pipeline (S2P) [33] is an automatic modular pipeline for push-broom images, taking stereo image pairs with RPCs as input and using the more global matching (MGM) stereo matching algorithm [34] to compute disparity.

In recent years, deep learning-based 3-D reconstruction methods have developed rapidly. Chen et al. [35] proposed a self-supervised stereo matching method for urban satellite images, first selecting reliable matching points using a superpixel-based random walk approach, followed by training a stereo matching network with a parallax-channel attention mechanism. Liang et al. [36] introduced a stereo matching method based on multilevel cost volume and multiscale feature constancy, enhancing the model's robustness through a two-stage fine-tuning process. He et al. [37] proposed an end-to-end hierarchical multiscale matching network for disparity estimation in high-resolution satellite stereo images. Xu et al. [38] introduced a stereo matching network based on bilateral grid learning, with an upsampling module for cost volumes that can be seamlessly integrated into many existing stereo matching networks for real-time processing. Several deep learning-based methods for 3-D reconstruction have dominated top rankings on mainstream datasets. However, traditional methods still offer effective solutions when processing larger or more numerous images, as deep learning methods are limited by sample quality and GPU memory requirements.

B. 3-D Reconstruction From Multiview Satellite Images

The 3-D reconstruction from multiview satellite imagery can effectively address challenges such as occlusion, shadows, and atmospheric noise by providing rich spatial redundancy. This section provides a brief overview of recent advances in satellite MVS reconstruction using commercial software, traditional methods, and deep learning methods.

Mainstream commercial software packages include MicMac [11], Agisoft Metashape [10], CATALYST [39], and ArcGIS [40]. Rupnik et al. [11] employed MicMac for stereo matching of high-resolution satellite imagery and utilized multidirectional dynamic programming for the fusion of multiview depth maps. Agisoft Metashape enables the generation of high-accuracy dense point clouds and DSMs from multiview satellite images with RPCs, supporting processing of various types of satellite imagery. CATALYST extracts 3-D information from multiview satellite images using a multiview SGM technique. ArcGIS generates DSMs using the Ortho Mapping module, suitable for terrain modeling and spatial analysis. Zhang et al. [18] approximated the RPC model as a perspective camera model and then adopted COLMAP for multiview satellite imagery 3-D reconstruction.

Traditional satellite MVS methods generally focus on two crucial steps: selecting high-quality image pairs and developing fusion algorithms for DSMs or point clouds. For pair selection, d'Angelo et al. [41] suggested that intersection angles between 15° and 25° are more suitable for urban and industrial areas. Qin [20] proposed a learning-based approach to identify high-quality stereo pairs. In this method, pairs with intersection angles ranging from 10° to 30° are ranked by minimizing the least-squares difference between the generated DSMs and ground-truth (GT) data. Facciolo et al. [9] analyzed factors most relevant to DSM quality using the IARPA MVS 3-D Mapping Challenge (MVS3D) dataset and proposed a simple heuristic method based on image metadata parameters. Pairs were first preselected with intersection angles between 5° and 45° and maximum off-nadir angles below 40° and then ranked based on their acquisition time intervals. Qin [22] proposed an optimized pair selection strategy by analyzing key meta-parameters of image pairs. This method additionally considered the effects of sun angle differences and matching completeness, using a trained support vector machine (SVM) model to predict stereo pair quality. Gomez et al. [17] introduced a simulation-based method to empirically map the relation between the orientation of the views and the reconstruction quality. A simulation tool was used to generate 3-D scene views from multiple directions and to precompute a map that encodes the geometric configuration of the views and their reconstruction quality. This map served as a proxy for the quality of real pairs. However, these methods do not consider the redundancy between stereo pairs and the completeness of spatial coverage. Such limitations may lead to uneven view distribution and concentrated occlusions, ultimately degrading the reconstruction quality in complex urban scenes.

DSM fusion is another critical step in traditional multiview 3-D reconstruction pipelines. Several fusion strategies have been proposed, typically involving WA, MF, and their variants, applied to each pixel's height value in a set of DSMs. While MF can remove outliers from a group of DSMs, it is highly sensitive to poor observations. WA, on the other hand, requires precise weight information, which may introduce biases. Qin [20] proposed an adaptive depth fusion (ADF)

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TABLE I
COMPARISON OF SATELLITE MVS RECONSTRUCTION METHODS

Method	Data Dependency	Computational Resources	Stereo Pair Selection	DSM Fusion
Commercial Software	Dependence on input image quality	Well-optimized processing workflow	Simple rule-based selection; lack of geometric diversity and coverage consideration	Basic fusion methods; susceptible to low-quality reconstruction effects
Traditional Methods	Low data requirement; strong generalization capability	Manageable computational cost; suitable for large-scale scenarios	Multi-parameter quality evaluation; lack of geometric diversity and coverage consideration	Support for auxiliary information inputs (e.g., uncertainty); customizable fusion strategies
Deep Learning	Dependence on large-scale, labeled training data; limited cross-domain generalization	High training cost; reliance on GPU and memory resources	Immature selection mechanisms; reliance on fixed input formats	Potential for learning-based fusion rules

method that considers spatial and color proximity to define irregular regions around each pixel, within which MF is applied to DSMs. This approach enhances the robustness of traditional MF while effectively preserving boundary information. In multidate imagery, vegetation changes may cause incoherent results. To address this multimodality in height values, Facciolo et al. [9] introduced a method that selects the mode corresponding to the ground altitude, which is the lowest one. K-medians clustering is used to estimate the height modes at each point, increasing the number of clusters until their span falls below a predefined precision threshold. Due to the impact of changing objects and occlusions, the median of a set of height values for a single pixel may no longer be the optimal assumption. To improve the DSM fusion process, Qin et al. [24] proposed an uncertainty-guided depth fusion (UGDF) method that adaptively guides the fusion process using auxiliary information derived from MVS algorithms. Perko and Zach [42] introduced a novel globally optimal robust DSM fusion method, defining a convex energy functional with a β-Lipschitz continuous gradient, thus simplifying the minimization problem. Gomez et al. [17] applied a bilateral filter framework in an iterative scheme, which allows the solution to be gradually refined. By using stricter height ranges, the focus can shift to height samples closer to the previous estimation, potentially improving accuracy. Effectively integrating the complementary information from multiple DSMs can improve reconstruction accuracy in urban scenes with dense occlusions and complex elevation variations.

Deep learning-based MVS methods have demonstrated impressive performance [43], [44], [45], [46], [47]. MVSNet [48] is one of the earliest end-to-end learnable solutions that infers depth maps from multiview images. Coarse-to-fine pyramid matching structures have also been introduced, such as CasMVSNet [13] and REDNet [15]. Stucker and Schindler [49] proposed a deep residual prior for 3-D reconstruction of high-resolution satellite images, which can resist variations in imaging conditions and acquisition geometry, enabling continuous optimization of DSMs. To adapt advanced learning-based MVS techniques for satellite MVS tasks, Gao et al. [8] proposed a universal deep learning framework for multiview satellite imagery 3-D reconstruction based on a differentiable RPC warping module. The RPCs are recorded as a tensor, and the RPC warping is formulated as a series of tensor transformations. Subsequently, Gao et al. [16] introduced a general deep learning-based framework, named SatMVSF, to perform 3-D reconstruction of the Earth's surface from multiview optical satellite images. The framework encompasses a complete processing pipeline and addresses the complexity and diversity of real-world scenes by incorporating a self-refinement strategy that enhances the performance and robustness of the SatMVSF framework without requiring any GT labels. Chen et al. [6] proposed a distribution contrast network for surface depth estimation from satellite MVS images, while Hong et al. [7] introduced a novel highly applicable iterative network for 3-D reconstruction from multiview satellite imagery. Both methods were primarily evaluated on the WHU-TLC dataset. However, existing deep learning-based MVS methods typically rely on end-to-end training frameworks and lack mature mechanisms for stereo pair selection, often operating with fixed input sets. In addition, they require large-scale, well-annotated training datasets and demand substantial GPU memory resources. A comparison of satellite MVS reconstruction methods is presented in Table I.

III. METHODOLOGY

This article proposes a novel satellite MVS method based on optimized image combination selection and confidence-guided DSM fusion. This section introduces the method in four aspects: optimized image combination selection, pair-wise stereo matching, confidence construction, and confidence-guided DSM fusion. First, multiview images are partitioned into several subintervals based on their imaging geometry. The interval boundaries are adjusted to ensure an even distribution of images across the subintervals. A hierarchical heuristic criterion is designed by integrating observational condition information, enabling the selection of a multiview image combination that provides comprehensive coverage of viewing angles while minimizing redundancy. Next, a stereo matching method based on iterative optimization of hierarchical GSC cost is employed to generate multiple DSMs. Confidence features are then constructed using auxiliary information generated during the MVS reconstruction process.

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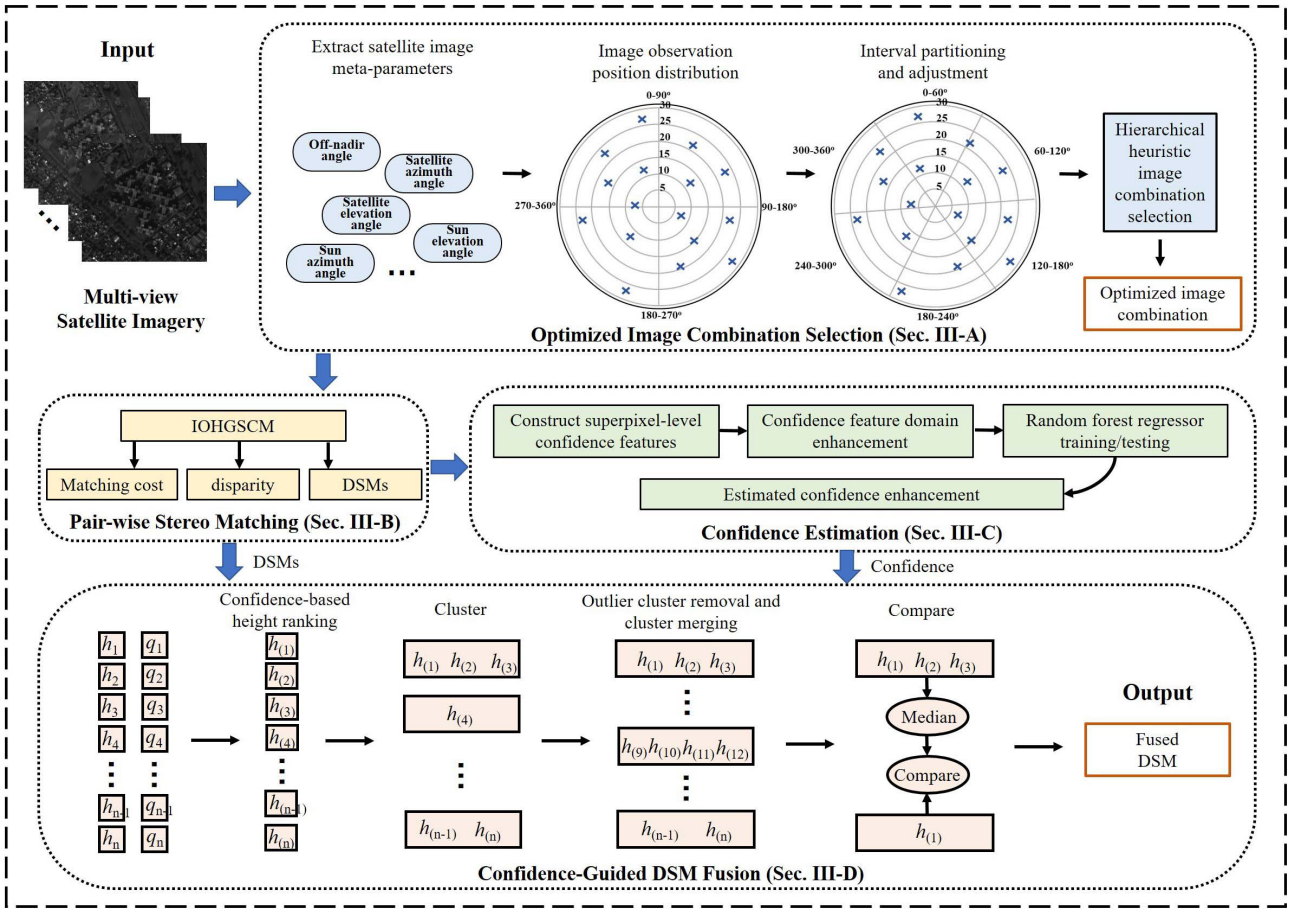


Fig. 1. Flowchart of the research methodology.

These confidence features and the estimated confidence are enhanced by incorporating contextual information. Finally, the constructed confidence is used to rank and cluster a set of height values. By balancing local statistical properties with single-point confidence, DSM fusion is achieved, producing more complete and accurate 3-D reconstruction results for urban areas. The overall workflow of the proposed satellite MVS method is illustrated in Fig. 1.

A. Optimized Image Combination Selection

This section focuses on partitioning multiview satellite images into several subintervals based on their imaging geometry. To ensure comprehensive coverage and low redundancy among multiview images, one image is selected from each interval to form an image combination. A hierarchical heuristic strategy is then designed to select an optimized image combination that satisfies conditions such as intersection angles between images, sun angle differences, and time differences in image acquisition. The technical pipeline of the optimized image combination selection is shown in Fig. 2.

The accuracy of traditional MVS method largely depends on the quality of stereo pairs, which is influenced by multiple factors, including intersection angles, sun angle differences, maximal off-nadir angles, and temporal proximity. For multiview image datasets, low-quality or degraded satellite images, such as those affected by snow cover, heavy fog, or cloud occlusion, can be preemptively removed. Additionally, images with the off-nadir angle of $\theta_{nadir} \geq 40^\circ$ can

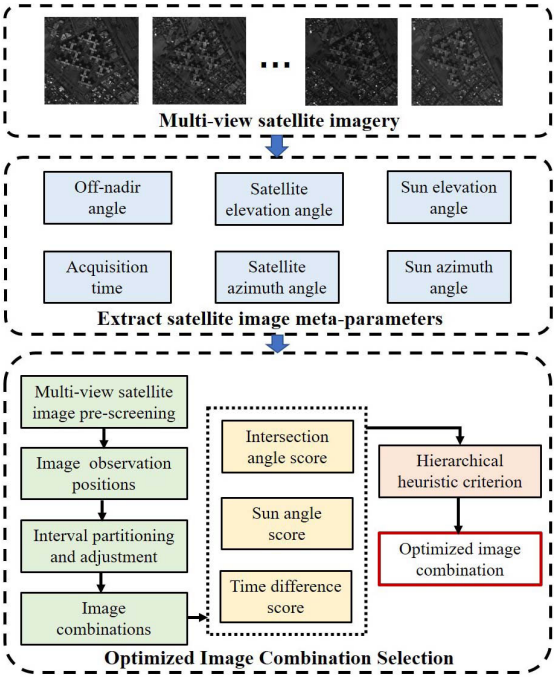


Fig. 2. Technical pipeline of the optimized image combination selection.

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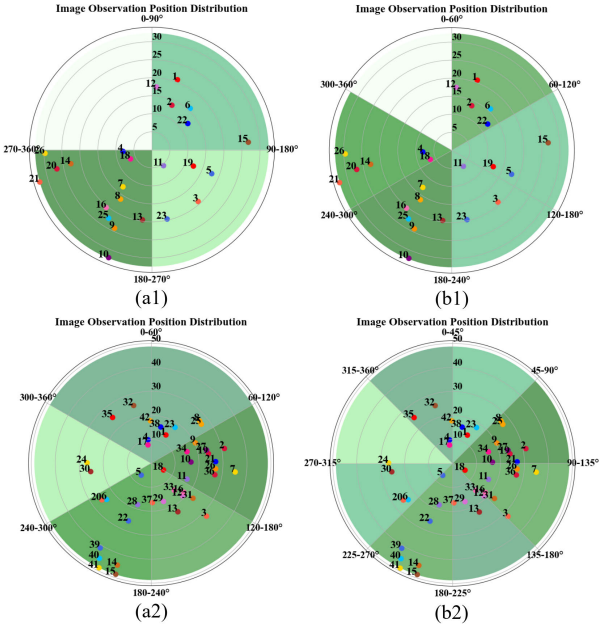


Fig. 3. Distribution of multiview image observation positions. US3D_JAX (a1) $N = 4$ and (b1) $N = 6$. MVS3D (a2) $N = 6$ and (b2) $N = 8$.

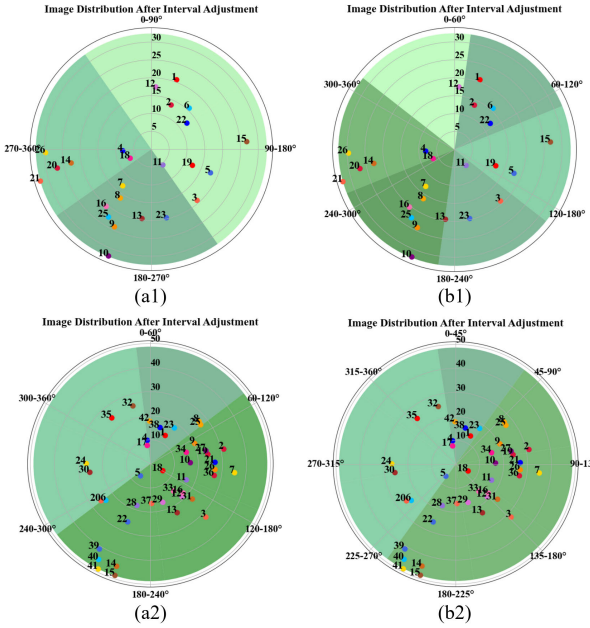


Fig. 4. Distribution of multiview images after interval adjustment. US3D_JAX (a1) $N = 4$ and (b1) $N = 6$. MVS3D (a2) $N = 6$ and (b2) $N = 8$.

be excluded. As noted in [22], a larger θ_{max_n} (maximal off-nadir angle of the two images) results in lower image resolution, more vertical surfaces, and a higher likelihood of occlusion and shadow effects during stereo matching. To ensure comprehensive angle coverage in multiview images, the observation positions of the images are plotted in a polar coordinate system based on their imaging geometry, and the multiview images are divided into N subintervals ($N \geq 3$). The satellite elevation angle and satellite azimuth angle are extracted to represent the observation positions. Using the US3D_JAX dataset and the MVS3D dataset as examples, the observation positions of the images are plotted in polar coordinates based on satellite elevation and azimuth angles. As shown in Fig. 3, the distribution of multiview images across different intervals is illustrated. In Fig. 3, different colors represent the number of images within each interval. Lighter colors indicate fewer images, while darker colors signify a higher concentration of images. It can be observed that some intervals have dense image distributions, while others are sparse or lack images entirely, which can negatively affect the subsequent image selection process. To address this issue, the interval boundaries are adjusted to distribute the images as evenly as possible across all intervals. By testing different starting angles $\theta_{start} \in [0^o, 360^o/N]$ and calculating the number of images in each interval $\{n_i\}_{i=1}^N$, the standard deviation of $\{n_i\}_{i=1}^N$ is computed to identify the configuration with the most uniform distribution. n_i denotes the number of images in the ith interval. The adjusted image distributions are shown in Fig. 4. The closer the colors across intervals, the more evenly distributed the images are. It is evident that the image distribution becomes more balanced after the interval adjustments. This balanced distribution ensures that the selected multiview image combinations provide comprehensive angle coverage while reducing redundancy, thereby laying the foundation for subsequent reconstruction steps.

In each interval, an image is arbitrarily selected to form a combination that ensures comprehensive coverage of observation angles. Subsequently, a hierarchical heuristic criterion is designed to identify the optimal multiview image combination. The selection principles consider four key factors: the intersection angle between images θ_{in}, the sun angle difference between images θ_{sun_dif}, the sun elevation angle of each image θ_{sun_el}, and the time difference between image acquisitions $time_{dif}$. The intersection angle between images θ_{in} affects the accuracy of 3-D reconstruction. A larger intersection angle often leads to greater occlusion of urban objects, reducing the overall accuracy of the DSM. Conversely, a smaller intersection angle generally ensures better completeness due to reduced occlusion but increases the uncertainty of 3-D points due to a lower base-to-height ratio. Following [9] and [20], an initial setting of $\theta_{in} \in [5, 35]$ is adopted, with further refinement to $\theta_{in} \in [15, 25]$ for more precise selection. The difference in sun angles θ_{sun_dif} consists of the difference in sun elevation angles θ_{sunel_dif} and the difference in sun azimuth angles θ_{sunaz_dif}, denoted as $\theta_{sun_dif} = (\theta_{sunel_dif}{}^2 + \theta_{sunaz_dif}{}^2)^{1/2}$. The sun elevation angle influences shadow length, while the sun azimuth angle determines shadow direction. Therefore, minimizing the sun angle difference between images is preferable. In addition to sun angle differences, a higher sun elevation angle θ_{sun_el} for individual images is desirable, as it reduces shadow interference in the reconstruction process. Finally, the time difference between image acquisitions $time_{dif}$ impacts the extent of ground surface changes and seasonal variations.

To select a multiview image combination that ensures comprehensive information coverage while minimizing redun-

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dancy, a hierarchical heuristic image selection strategy is designed as follows.

First, the intersection angle score of the multiview image combination is computed. Using satellite image meta-parameters, the intersection angle between images is derived from the satellite elevation angle and satellite azimuth angle as follows:

$$c_{ij} = \sin\left(\theta^i_{sat_el}\right)\cdot\sin\left(\theta^j_{sat_el}\right) + \cos\left(\theta^i_{sat_el}\right) \cdot\cos\left(\theta^j_{sat_el}\right)\cdot\cos\left(\theta^i_{sat_az} - \theta^j_{sat_az}\right) \quad (1)$$

$$\theta_{in} = \frac{180}{\pi}\arccos\left(c_{ij}\right) \quad (2)$$

where $\theta^i_{sat_az}, \theta^i_{sat_el}, \theta^j_{sat_az}$, and $\theta^j_{sat_el}$ represent the satellite azimuth and elevation angles of the ith and jth images, respectively, and arccos(·) denotes the inverse cosine function. The intersection angle score for the multiview image combination is defined as

$$score_{\theta_{in}} = \omega_{in1}\cdot\text{count}_{5_35}\left(\{\theta_{in}\}\right) + \omega_{in2}\cdot\text{count}_{15_25}\left(\{\theta_{in}\}\right) \quad (3)$$

where $score_{\theta_{in}}$ represents the score for the intersection angle, $\text{count}_{5_35}(\cdot)$ denotes the number of image pairs satisfying $\theta_{in} \in [5, 35]$ within the intersection angle set $\{\theta_{in}\}$, and $\text{count}_{15_25}(\cdot)$ represents the number of image pairs satisfying $\theta_{in} \in [15, 25]$ within $\{\theta_{in}\}$. The total number of image pairs is $N\cdot(N-1)/2$, where N refers to the number of intervals, and ω_{in1} and ω_{in2} are weight parameters.

Next, the sun angle score of the multiview image combination is computed. The difference in sun angles is determined using the sun elevation and azimuth angles from the satellite image meta-parameters

$$\theta_{sun_dif} = \sqrt{\left(\theta^i_{sun_az} - \theta^j_{sun_az}\right)^2 + \left(\theta^i_{sun_el} - \theta^j_{sun_el}\right)^2} \quad (4)$$

where $\theta^i_{sun_az}, \theta^i_{sun_el}, \theta^j_{sun_az}$, and $\theta^j_{sun_el}$ represent the sun azimuth and elevation angles of the ith and jth images, respectively. Additionally, the sun elevation angle θ_{sun_el} of an individual image is a crucial factor in evaluating the quality of an image combination. Ideally, the sun angle difference between image pairs should be minimized, while the average sun elevation angle across the multiview image combination should be maximized. The sun angle score for the multiview image combination is defined as

$$score_{\theta_{sun}} = \frac{2\cdot\omega_{sun_dif}}{N\cdot(N-1)}\cdot\sum_{n=1}^{N\cdot(N-1)/2}\theta^n_{sun_dif} + \frac{\omega_{sun_el}}{N}\cdot\sum_{m=1}^{N}\left(90 - \theta^m_{sun_el}\right) \quad (5)$$

where $score_{\theta_{sun}}$ represents the sun angle score of the image combination; ω_{sun_dif} and ω_{sun_el} are weight parameters corresponding to the sun angle difference and the sun elevation angle, respectively; n denotes the nth stereo image pair; and m denotes the mth image.

Then, the time difference score of the multiview image combination is computed. As noted in [9], stereo image pairs acquired in the same season across different years can still yield high-quality results. Therefore, the time difference between image acquisitions $time_{dif}$ is defined as

$$days_{dif} = \left|\text{DOY}\left(t_1\right) - \text{DOY}\left(t_2\right)\right| \quad (6)$$

$$time_{dif} = \begin{cases} days_{dif}, & days_{dif} \le 182 \\ 365 - days_{dif}, & 182 < days_{dif} < 365 \end{cases} \quad (7)$$

where t_1 and t_2 represent the acquisition times of two images and DOY(t) gives the day of year of time t within its calendar year. The time difference score for the multiview image combination is defined as

$$score_{time} = \frac{2}{N\cdot(N-1)}\cdot\sum_{n=1}^{N\cdot(N-1)/2} time^n_{dif} \quad (8)$$

where $score_{time}$ represents the acquisition time score of the image combination.

A larger $score_{\theta_{in}}$ indicates a higher proportion of image pairs with favorable intersection angles, which is desirable. A smaller $score_{\theta_{sun}}$ suggests that the sun angle differences are minimized and the sun elevation angles are higher, reducing the impact of shadow variations on stereo matching. A smaller $score_{time}$ implies a lower time difference between image pairs, thereby minimizing the influence of surface changes and seasonal variations. Among these factors, the intersection angle is the most critical, as it directly determines the accuracy of 3-D information retrieval. The sun angle is the second most important factor, as it affects shadow direction and length, influencing stereo matching precision. The time difference is relatively less critical, as changes in surface features can be partially mitigated through stereo matching algorithms. Based on these considerations, a hierarchical heuristic criterion is designed to optimize multiview image selection. The detailed steps are as follows.

Step 1: Select one image from each of the N ($N \ge 3$) subintervals to form N_{mul} multiview image combinations $\{C_m\}_{m=1}^{N_{mul}}$, where each combination $C = \{I_1, I_2, \ldots, I_N\}$ consists of $N\cdot(N-1)/2$ image pairs.

Step 2: Calculate $score_{\theta_{in}}$, $score_{\theta_{sun}}$, and $score_{time}$ for each image combination, denoted as $\{score^m_{\theta_{in}}\}_{m=1}^{N_{mul}}$, $\{score^m_{\theta_{sun}}\}_{m=1}^{N_{mul}}$, and $\{score^m_{time}\}_{m=1}^{N_{mul}}$.

Step 3: Rank the multiview image combinations in descending order based on $score_{\theta_{in}}$, and select the top N_1 combinations, denoted as $\{C^m_{sort^{\downarrow}_{in}}\}_{m=1}^{N_1}$.

Step 4: Sort the selected N_1 combinations in ascending order based on $score_{\theta_{sun}}$ and $score_{time}$, yielding two ranked sequences $\{C^m_{sort^{\uparrow}_{sun}}\}_{m=1}^{N_1}$ and $\{C^m_{sort^{\uparrow}_{time}}\}_{m=1}^{N_1}$. The intersection of the top N_2 combinations from both sequences is computed, resulting in $\hat{C} = \{C^m_{sort^{\uparrow}_{sun}}\}_{m=1}^{N_2} \cap \{C^m_{sort^{\uparrow}_{time}}\}_{m=1}^{N_2}$.

Step 5: Determine the optimal multiview image combination based on the following.

1) If $\hat{C}$ is empty, the optimal multiview image combination $\hat{C}_{opt}$ is the one in $\{C^m_{sort^{\downarrow}_{in}}\}_{m=1}^{N_1}$ with the smallest sun angle score, denoted as $\hat{C}_{opt} = C^1_{sort^{\uparrow}_{sun}}$.
2) If $\hat{C}$ is not empty and satisfies $\text{count}(\hat{C}) = 1$, the optimal multiview image combination $\hat{C}_{opt}$ is $\hat{C}$, denoted as $\hat{C}_{opt} = \hat{C}$.

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3) If $\hat{C}$ is not empty and satisfies count($\hat{C}$) $\neq 1$, the optimal multiview image combination $\hat{C}_{opt}$ is the one in $\hat{C}$ with the smallest sun angle score, denoted as $\hat{C}_{opt} = \hat{C}_{sun_{min}}$.

Step 6: The selected optimal multiview image combinations $\{\hat{C}_{opt}\}$ from N ($N \geq 3$) subintervals are validated by generating a DSM over a small test region and comparing it with GT data. This process determines the optimal number of subintervals.

In summary, this method effectively optimizes the selection of multiview image combinations, ensuring comprehensive information coverage while minimizing redundancy. The optimized image combinations achieve superior performance in terms of intersection angles, sun angle differences, sun elevation angles, and time acquisition differences. These improvements provide high-quality data support for subsequent DSM generation and fusion processes.

B. Pair-Wise Stereo Matching

The pair-wise stereo matching method proposed in our previous work [29] is employed in this study. To enhance robustness against illumination variations and small ground surface changes, a GSC cost is utilized, which is defined as follows:

$$\begin{aligned} C^{p_r}_{(m,n')} = \frac{\sigma_g}{K}\sum_{k=1}^{K}\Big| & d_g\left(p_r\left(m,n'\right), p_r\left[\left(i'',j''\right)^k\right]\right) \\ & - d_g\left(p_r\left(m,n'\right), p_r\left[\left(i',j'\right)^k\right]\right)\Big| \\ & + \frac{\sigma_c}{S}\sum_{s=1}^{S}\Big| H\left(p_r\left(m,n'\right), p_r\left[\left(i'',j''\right)^s\right]\right) \\ & - H\left(p_r\left(m,n'\right), p_r\left[\left(i',j'\right)^s\right]\right)\Big| \end{aligned} \tag{9}$$

where $d_g(p_r(m,n'), p_r(i'',j''))$ represents the grayscale spatial distance measurement, $H(p_r(m,n'), p_r(i'',j''))$ represents the spatial relative relationship measurement, and σ_g and σ_c are the weights corresponding to the two measurement methods.

The matching cost is formulated as a truncated weighted combination of the GSC cost and gradient cost. Additionally, multiscale weighted aggregation is applied to the constructed matching cost.

An object-based iterative optimization is employed to continuously detect occlusions and refine boundary disparities through iterative visibility term and disparity discontinuity term. The matching cost is iteratively updated by the proposed algorithm as follows:

$$\mathrm{C}^d_{t+1} = \overline{\mathrm{W}}\left((1-\lambda)\,\mathrm{V}^d_t + \lambda \boldsymbol{\Psi}^d_t\right) \tag{10}$$

where C^d_{t+1} represents the updated matching cost, W denotes the weight matrix, $\boldsymbol{\Psi}^d_t = [\psi_t(d,d')]_{\,k'\times \max d}$ represents the disparity discontinuity term of the tth iteration, $\mathrm{V}^d_t = C^d_t \odot [O_t(s)]_{\,k'\times 1}$ denotes the visibility term of the tth iteration, k' represents the number of superpixels, $\max d$ indicates the maximum disparity value, and λ is used to balance the visibility term and the disparity discontinuity term.

C. Confidence Estimation

Auxiliary information generated by the pair-wise stereo matching algorithm can be leveraged to describe matching errors and reconstruction accuracy, thereby effectively guiding the DSM fusion process. Inspired by [24], [50], and [51], multiple confidence measures are computed using the matching costs and disparity values obtained from the pair-wise stereo matching process. Influential confidence features are then selected from the candidates to enhance the analysis of matching errors in remote sensing data. Since the pair-wise stereo matching algorithm employed in this study adopts an object-based iterative optimization strategy, the constructed confidence features are also object-based and exhibit robustness against outliers. To further improve the ability of confidence measures to characterize matching errors, contextual information is incorporated to enhance the confidence features and the estimated confidence.

First, high-dimensional filtering is applied to superpixel-level confidence features using an adaptive manifold filter, formulated as follows:

$$\hat{\mathrm{f}}_u = \sum_{v\in\mathcal{N}_u} w_{u,v}\,(\mathrm{f})\mathrm{f}_v \tag{11}$$

where f_v represents the confidence feature vectors of neighboring superpixels, $\hat{\mathrm{f}}_u$ denotes the enhanced confidence feature vector of the target superpixel, and $w_{u,v}(\mathrm{f})$ is the weight function based on confidence feature similarity, defined as

$$w_{u,v}\,(\mathrm{f}) = \frac{\exp\left(-\left\|\mathrm{f}_u - \mathrm{f}_v\right\|^2/\sigma_{\mathrm{f}}\right)}{\sum_{k\in\mathcal{N}_u}\exp\left(-\left\|\mathrm{f}_u - \mathrm{f}_k\right\|^2/\sigma_s\right)} \tag{12}$$

where σ_f is the standard deviation of the Gaussian function in the feature domain. The neighborhood range $\mathcal{N}_u$ is adaptively determined by σ_s, which represents the standard deviation of the Gaussian function in the spatial domain. Confidence features of neighboring superpixels are adaptively aggregated based on their similarity to the target superpixel to enhance the confidence representation.

A random forest regressor is then trained on the enhanced confidence features and GT confidence values to generate confidence estimates, denoted by Q^*. To further improve the confidence map quality, the estimated confidence values are refined using a K-nearest neighbor (K-NN) filtering approach, formulated as

$$\hat{Q}_u = \sum_{v'\in\mathcal{N}^K_u} Q^*_{v'}/K \tag{13}$$

where $\mathcal{N}^K_u$ represents K-NN for superpixel u, K is the number of neighboring superpixels, $Q^*_{v'}$ denotes the estimated confidence value of the neighboring superpixel v', and $\hat{Q}_u$ denotes the enhanced confidence value of superpixel u. The neighbors are selected based on feature-space similarity to superpixel u. Theoretically, the more similar the confidence features, the closer the estimated confidence values.

By leveraging auxiliary information from the pair-wise stereo matching process for confidence estimation and incorporating contextual information to enhance the confidence features and estimated confidence, the resulting confidence exhibits strong robustness and can effectively guide the DSM fusion process.

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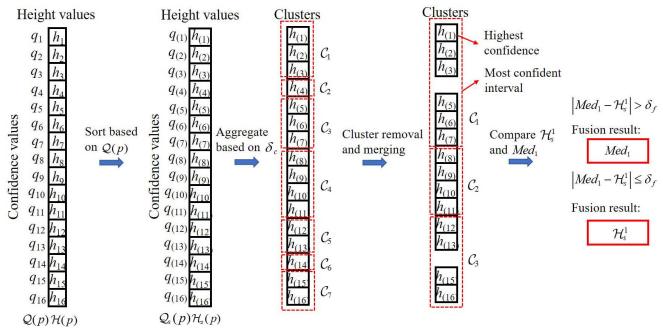


Fig. 5. Confidence-guided DSM fusion process.

D. Confidence-Guided DSM Fusion

DSM fusion is a key technique for improving the quality of 3-D reconstruction in urban areas. This study proposes a confidence-guided DSM fusion method, where confidence information is incorporated to rank and cluster height values. By integrating the statistical properties of local regions with single-point confidence, an effective DSM fusion strategy is developed. The proposed method effectively suppresses outlier interference, enhances adaptability to complex height distributions, and enables robust DSM fusion.

Multiple DSMs are generated by pairing selected multiview images and processing them using the pair-wise stereo reconstruction algorithm. These DSMs are then projected into a unified geographic coordinate system for alignment and fusion. A straightforward approach would be to take the median of the aggregated height samples from all DSMs. However, large intersection angles in stereo image pairs often introduce errors in structures with significant elevation variations. Consequently, selecting the median height may not reliably represent the true elevation at a given location. To address this issue, a confidence-guided DSM fusion method is proposed. Fig. 5 visualizes the confidence-guided DSM fusion process.

First, the multiple height values $\mathcal{H}(p) = \{h_n(p), n = 1, 2, \ldots, M\}$ corresponding to each point p are sorted in descending order based on their confidence values $\mathcal{Q}(p) = \{q_n(p), n = 1, 2, \ldots, M\}$, generating

$$\left[\mathcal{H}_s(p), \mathcal{Q}_s(p)\right] = \left\{\left(h_{(n)}(p), q_{(n)}(p)\right), n = 1, 2, \ldots, M\right\} \quad (14)$$

where $q_n(p)$ represents the confidence of the nth height value for point p and M denotes the total number of DSMs.

Next, based on the ranking order of the height samples, height values satisfying the threshold δ_c are grouped into the same cluster. A height value is included in the current cluster $\mathcal{C}$ if the following condition is met:

$$\max(\mathcal{C}) - \min(\mathcal{C}) \le \delta_c. \quad (15)$$

Otherwise, a new cluster is initiated, with the initial clustering result denoted as $\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_k\}$.

To dynamically adjust the threshold δ_c based on the distribution characteristics of height values, the following formulation is employed:

$$\delta_c = k_c \cdot \sigma_c$$

$$\sigma_c = \sqrt{\frac{1}{n_c}\sum_{i=1}^{n_c}(h_i - \mu_c)^2}$$

$$\mu_c = \frac{1}{n_c}\sum_{i=1}^{n_c} h_i \quad (16)$$

where n_c represents the number of height values in the current cluster, k_c is an adjustment factor, μ_c is the mean height within the cluster, and σ_c is the standard deviation of the height distribution.

Due to potential inaccuracies in confidence estimation, outlier clusters may arise within the sorted height sequence, fragmenting height values that should belong to the same cluster. Therefore, it is necessary to identify anomalous clusters and merge clusters that meet specific conditions.

A cluster $\mathcal{C}_j$ is considered an outlier if the number of points it contains is below a predefined threshold minPts and if its height mean differs significantly from both adjacent clusters, failing to meet the threshold δ_m

$$|\mathcal{C}_j| < \text{minPts} \,\&\, |\mu_{\mathcal{C}_j} - \mu_{\mathcal{C}_{j-1}}| > \delta_m \,\&\, |\mu_{\mathcal{C}_j} - \mu_{\mathcal{C}_{j+1}}| > \delta_m \,\&\, |\mu_{\mathcal{C}_{j-1}} - \mu_{\mathcal{C}_{j+1}}| \le \delta_m \quad (17)$$

$$\delta_m = k_m \cdot \delta_c \quad (18)$$

where $\mu_{\mathcal{C}_j}$ represents the mean height value of cluster $\mathcal{C}_j$; $\mu_{\mathcal{C}_{j-1}}$ and $\mu_{\mathcal{C}_{j+1}}$ denote the mean height values of the neighboring clusters $\mathcal{C}_{j-1}$ and $\mathcal{C}_{j+1}$, respectively; and k_m is an adjustment factor.

Identified outlier clusters $\mathcal{C}_j$ are removed, and adjacent clusters $\mathcal{C}_{j'}$ and $\mathcal{C}_{j'+1}$ are further examined for potential merging based on the following condition:

$$\left|\mu_{\mathcal{C}_{j'}} - \mu_{\mathcal{C}_{j'+1}}\right| \le \delta_m. \quad (19)$$

δ_m is dynamically adjusted based on intracluster dispersion, enforcing stricter merging conditions for more compact clusters. If the merging criterion is satisfied, the two clusters are combined into a new cluster $\mathcal{C}_{new}$

$$\mathcal{C}_{new} = \mathcal{C}_{j'} \cup \mathcal{C}_{j'+1}. \quad (20)$$

Additionally, if the clusters at the extremes of a set of height values contain fewer than minPts points, they are merged with their adjacent clusters. The final output consists of the refined clustering results after removing anomalous clusters and merging adjacent clusters, denoted as $\{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_K\}$.

Finally, DSM fusion is performed based on the height clustering results obtained through confidence ranking. The height values in $\mathcal{H}_s(p)$ are divided into K groups, denoted as $\mathcal{H}_{rs}^k(p)$. The median height value Med_K is then computed for each group

$$Med_K(p) = \left\{\text{Median}\left(\cup_{k=1,\ldots,K}\mathcal{H}_{rs}^k(p)\right)\right\}. \quad (21)$$

Intuitively, as K increases, each group is likely to contain a growing number of erroneous values. The first median value, $Med_1(p)$, corresponds to the most confident clustering interval, capturing local statistical characteristics. Given the robust confidence estimation method employed in this study, the height value $\mathcal{H}_s^1(p)$ associated with the highest confidence $\mathcal{Q}_s^1(p)$ should theoretically have the smallest reconstruction

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error. However, to account for potential confidence estimation errors, $Med_1(p)$ is used to validate the reliability of $\mathcal{H}_s^1(p)$.

The height value with the highest confidence $\mathcal{H}_s^1(p)$ is compared against the median of the most confident height clustering interval $Med_1(p)$

$$H_{final}(p) = \begin{cases} Med_1(p), & \text{if } \left|Med_1(p) - \mathcal{H}_s^1(p)\right| > \delta_f \\ \mathcal{H}_s^1(p), & \text{if } \left|Med_1(p) - \mathcal{H}_s^1(p)\right| \le \delta_f \end{cases} \quad (22)$$

where δ_f is a threshold parameter. If the difference between $Med_1(p)$ and $\mathcal{H}_s^1(p)$ is smaller than δ_f, $\mathcal{H}_s^1(p)$ is considered reliable and selected as the final fused height. Otherwise, $Med_1(p)$ is chosen as the final fused height, preserving the statistical characteristics of the local high-confidence region.

Following these steps, multiple DSMs are fused on a point-by-point basis to generate the final DSM fusion result. The proposed confidence-guided DSM fusion method effectively integrates both single-point reliability and local saliency of multiple height values. It provides a more accurate representation of ground elevations while enhancing robustness to outliers and improving the capture of fine local details.

IV. Experiments

In this section, qualitative and quantitative experiments are carried out on multiview satellite imagery. Section IV-A introduces the experimental data and evaluation metrics, analyzes the experimental parameters and multiview image selection strategies, evaluates the estimated confidence and DSM fusion method, and compares the end-to-end performance of the proposed satellite MVS method with mainstream methods.

A. Experimental Data and Evaluation Metrics

The methods described above were tested on two datasets, comprising stereo satellite images from the MVS3D dataset [52] and the US3D dataset [53].

The MVS3D dataset consists of 47 WorldView-3 images covering a 100-km^2 area near San Fernando, Argentina. These images were acquired over a 14-month period, with most taken on different dates. The dataset exhibits variations in image quality, with winter images being considerably noisier. Additionally, images captured at large incidence angles suffer from a loss of resolution in the range direction. Airborne LiDAR data for the same area is provided and is considered the GT for elevation.

The US3D dataset includes 26 WorldView-3 images acquired between 2014 and 2016 over Jacksonville (JAX), Florida, and 43 WorldView-3 images collected between 2014 and 2015 over Omaha (OMA), Nebraska. Semantic labels and airborne LiDAR data are also available, with the LiDAR data serving as the GT for elevation.

To assess the quality of the reconstruction results, the following evaluation metrics [23] are employed.

1) *Mean Absolute Error (MAE):* The average of the L_1 distance between the GT and the predicted DSM.
2) *Root-Mean-Square Error (RMSE):* The standard deviation of the residuals between the GT and the predicted DSM.
3) *Median Height Error (MHE):* The median value of the absolute error between the GT and the predicted DSM.
4) *Completeness (COMP):* Proportion of evaluated pixels where the altitude of the predicted DSM differs from the GT less or equal to 1 m.
5) *Time:* The run time required for the entire process from importing multiview satellite images to obtaining the output DSM.

B. Experimental Parameter Analysis

This section describes the parameter settings of the proposed satellite MVS method and provides an analysis of key parameters. The parameters used in this study are derived from experimental analysis, prior research, and empirical values. The parameters for optimized image combination selection are primarily determined through experimental analysis, while those for pair-wise stereo matching are adopted from our previous work [29]. The confidence estimation parameters are mainly based on empirical values, referencing [50]. The threshold parameters in confidence-guided DSM fusion are adaptively estimated.

An experimental analysis was conducted on key parameters involved in the optimized image combination selection process, including the number of intervals N and the weight parameters $\omega_{in1}, \omega_{in2}, \omega_{sun_dif}$, and ω_{sun_el} used in computing the intersection angle and sun angle scores. Multiview images from the US3D_JAX, US3D_OMA, and MVS3D datasets are divided into multiple subintervals. The number of subintervals N directly determines the number of images in the optimized combination. Fig. 6 illustrates the relationship between the number of intervals N and reconstruction accuracy. For the US3D_JAX and US3D_OMA datasets, increasing N significantly improves reconstruction performance, with the evaluation metrics rising rapidly up to $N = 6$, then progressing more slowly with minor fluctuations, and reaching their best values around $N = 8$. For the MVS3D dataset, reconstruction accuracy increases steeply up to $N = 8$, with only marginal gains beyond this point. These dataset-specific inflection points suggest that MVS3D is more challenging due to its complex urban structures, which introduce greater occlusion. Balancing computational efficiency against reconstruction quality, we recommend setting $N = 8$ as a practical default across datasets.

Parameters ω_{in1} and ω_{in2} weight the valid interval $[5°, 35°]$ and optimal interval $[15°, 25°]$ intersection angle scores, respectively. The valid interval indicates the basic geometric feasibility for stereo reconstruction, while the optimal interval represents the range in which stereo pairs are most likely to yield high geometric accuracy and stable matching performance. Parameters ω_{sun_dif} and ω_{sun_el} weight the sun angle difference score and the sun elevation score, respectively. For the sun angle score, both the average sun angle difference between image pairs and the average sun elevation angle across the image combination were considered to identify combinations with minimal shadow variation and short shadow lengths. The influence of these weights on reconstruction was evaluated experimentally. Set $\omega_{in1} \in [0.1, 0.9]$ ($\omega_{in2} = 1 - \omega_{in1}$) with a step size of 0.1, and set $\omega_{sun_dif} \in [0.1, 0.9]$ ($\omega_{sun_el} = 1 - \omega_{sun_dif}$) with a step size of 0.1. Fig. 7 illustrates the

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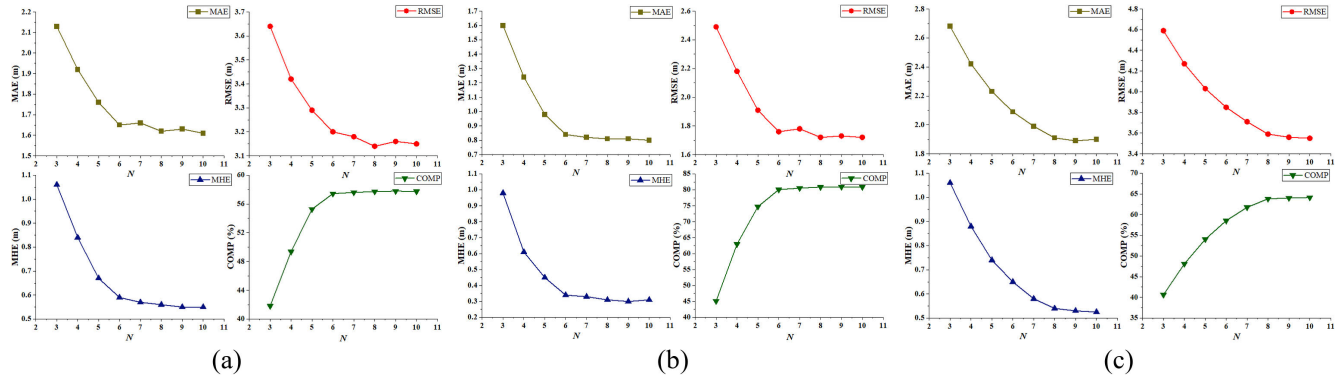


Fig. 6. Analysis of the number of intervals N. (a) US3D_JAX, (b) US3D_OMA, and (c) MVS3D.

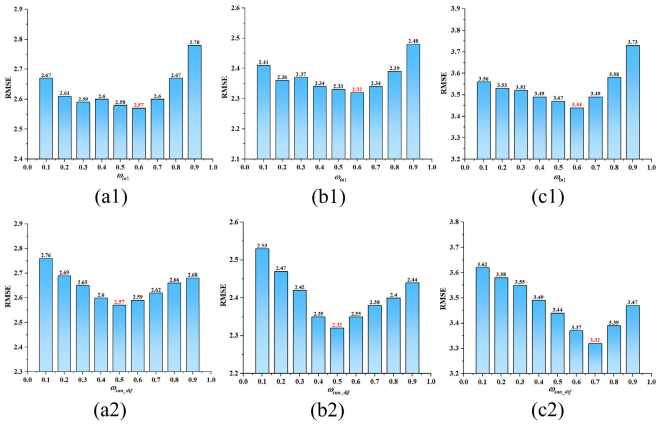


Fig. 7. Analysis of weight parameters ω_{in1} and ω_{sun_dif}. (a1) and (a2) US3D_JAX, (b1) and (b2) US3D_OMA, and (c1) and (c2) MVS3D.

impact of varying values of ω_{in1} and ω_{sun_dif} on reconstruction accuracy. Across multiple datasets, as ω_{in1} increases, the RMSE initially decreases slightly with fluctuations and then increases more rapidly. In contrast, increasing ω_{sun_dif} leads to a sharper initial decline in RMSE, followed by a relatively slower upward trend. On the US3D dataset, the best performance was obtained at $\omega_{in1} = 0.6$ and $\omega_{sun_dif} = 0.5$. On the MVS3D dataset, the best performance was obtained at $\omega_{in1} = 0.6$ and $\omega_{sun_dif} = 0.7$. The differing optimal values of ω_{sun_dif} suggest marked interdataset variation in sun angles, with sun angle difference exerting a stronger influence on MVS3D than on US3D. These results indicate that setting the intersection angle within an optimal interval $[15°, 25°]$ improves reconstruction accuracy and that sun angle difference is a critical factor influencing reconstruction quality. To provide a unified set of weights applicable across datasets, we examined how varying the weights affects reconstruction quality and recommend $\omega_{sun_dif} = 0.6$ and $\omega_{sun_el} = 0.4$, which yield near-optimal performance on all datasets. Consequently, we adopt a fixed cross-dataset configuration of $N = 8$, $\omega_{in1} = 0.6, \omega_{in2} = 0.4, \omega_{sun_dif} = 0.6$, and $\omega_{sun_el} = 0.4$ and use this setting consistently throughout all experiments in this article.

TABLE II
ABLATION STUDY ON OPTIMIZED IMAGE COMBINATION SELECTION

	Intersection angle	Sun angle	Time difference
AS1	✓		
AS2	✓	✓	
AS3	✓		✓
AS4	✓	✓	✓

TABLE III
QUANTITATIVE EVALUATION OF OPTIMIZED IMAGE COMBINATION SELECTION ABLATION RESULTS ACROSS MULTIPLE DATASETS

Dataset		AS1	AS2	AS3	AS4
US3D JAX	MAE(m)	1.58	1.52	1.54	1.43
	RMSE(m)	2.83	2.71	2.78	2.62
	MHE(m)	0.66	0.56	0.60	0.49
	COMP(%)	56.34	60.83	58.71	63.38
US3D OMA	MAE(m)	0.91	0.84	0.87	0.79
	RMSE(m)	1.85	1.74	1.79	1.68
	MHE(m)	0.40	0.31	0.34	0.27
	COMP(%)	72.42	79.05	75.84	82.81
MVS3D	MAE(m)	1.98	1.82	1.91	1.74
	RMSE(m)	3.52	3.33	3.40	3.23
	MHE(m)	0.76	0.58	0.68	0.50
	COMP(%)	54.59	61.97	57.92	64.95

C. Ablation Study on Optimized Image Combination Selection

To validate the effectiveness of the proposed optimized image combination selection method, ablation studies (ASs) were conducted to evaluate the influence of intersection angle, sun angle, and image acquisition time on image selection quality. Since the intersection angle is recognized as the most critical metadata parameter in the image selection process, the effects of sun angle and image acquisition time were analyzed under a fixed intersection angle setting. The ASs are summarized in Table II, covering three settings: 1) considering only the sun angle; 2) considering only the image acquisition time difference; and 3) jointly considering both factors. The corresponding quantitative results are presented in Table III. As shown in Table III, both sun angle and image acquisition time difference individually contribute to improved image selection quality across multiple datasets. Notably,

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TABLE IV
QUANTITATIVE EVALUATION OF IMAGE SELECTION STRATEGIES ACROSS MULTIPLE DATASETS

Data/ Method	Our				SHM [9]				LMPSM [22]				ST [17]			
	MAE (m)	RMSE (m)	MHE (m)	COMP (%)	MAE (m)	RMSE (m)	MHE (m)	COMP (%)	MAE (m)	RMSE (m)	MHE (m)	COMP (%)	MAE (m)	RMSE (m)	MHE (m)	COMP (%)
US3D JAX	1.43	2.62	0.49	63.38	1.65	2.94	0.71	54.93	**1.61**	**2.83**	**0.65**	**56.72**	1.56	2.78	0.62	58.34
US3D OMA	0.79	1.68	0.27	82.81	0.96	1.95	0.43	70.42	**0.89**	**1.83**	**0.37**	**73.79**	0.86	1.79	0.35	76.14
MVS3D	1.74	3.23	0.50	64.95	2.01	3.60	0.78	53.67	**1.92**	3.39	**0.71**	**56.95**	1.88	**3.43**	0.66	59.56

sun angle has a more pronounced effect on reconstruction accuracy, indicating that it is the second most important metadata parameter after intersection angle. On the MVS3D dataset, the improvement brought by incorporating sun angle is particularly significant. Furthermore, jointly considering both sun angle and image acquisition time difference yields superior reconstruction performance compared to considering either factor alone. Therefore, an image combination selection strategy that integrates intersection angle, sun angle, and image acquisition time difference supports the efficient selection of high-quality image sets and contributes to improved 3-D modeling accuracy in urban scenes.

D. Analysis of Image Selection Strategies

This study proposes an optimized multiview image selection method that leverages image meta-parameters and employs a hierarchical heuristic approach to select an image combination with balanced coverage and minimal redundancy. The selected images are paired to form multiple stereo image pairs. To assess the method's effectiveness, we compare it with established pairing strategies, including SHM [9], LMPSM [22], and ST [17].

Table IV provides a quantitative evaluation of various image selection strategies across different datasets. The red figures rank first, the blue figures rank second, and the bold figures rank third in each metric for each dataset. RMSE and MAE represent reconstruction accuracy, where lower values indicate higher accuracy. MHE reflects reconstruction stability, with lower values indicating greater stability. COMP measures reconstruction completeness, where higher values indicate a more complete reconstruction. A control variate method was adopted to ensure consistency in the DSM fusion algorithm and to enable a fair comparison. As shown in Table IV, the proposed optimized image combination selection strategy achieves the best reconstruction performance. ST [17] demonstrates strong performance by leveraging a simulation tool to generate synthetic images from multiple viewpoints and pre-establishing the relationship between reconstruction quality and geometric imaging parameters. LMPSM [22], which employs an SVM model to establish a relationship between reconstruction quality and image meta-parameters, exhibits slightly lower performance than ST [17]. Both methods rely on predefining the relationship between image meta-parameters and reconstruction quality. However, the effectiveness of multiview reconstruction depends not only on the quality of individual stereo image pairs but also on the completeness of the information coverage provided by the selected image pairs. Consequently, the proposed optimized image selection strategy achieves superior reconstruction performance. In contrast, SHM [9] employs a simple heuristic approach to select stereo image pairs, without considering factors such as variations in sun angles. Differences in shadow length and direction may introduce additional matching errors, resulting in inferior performance compared to other image selection strategies.

The experimental results quantitatively demonstrate that the proposed optimized image selection strategy consistently delivers outstanding performance across various urban scenarios.

E. Analysis of Confidence

In this study, a confidence-guided DSM fusion strategy is proposed, where the ability of estimated confidence to describe reconstruction accuracy is critically evaluated. To avoid overreliance on estimated confidence values, the proposed fusion strategy incorporates the local statistical properties of a set of height values. The effectiveness of the adopted confidence is demonstrated through visualized experimental results.

To assess the predictive ability of estimated confidence, confidence measure combinations, and individual confidence measures in reconstruction accuracy, sparsification curves are analyzed. The sparsification rate is defined by ranking reconstructed points in descending order of confidence and discarding the lowest 10%–100% of estimates. Completeness (COMP) denotes the metric of reconstruction completeness obtained at each sparsification rate by comparing the retained high-confidence estimates with the reference data. Accordingly, a completeness-sparsification curve is plotted to assess the effectiveness of the confidence estimates. A more pronounced upward trend in COMP indicates a stronger ability of the confidence measure to represent reconstruction accuracy. This study compares 22 commonly used confidence measures (denoted as C22) [54], a selected subset of eight confidence measures (C8), an enhanced version of C8 incorporating confidence feature domains and estimated confidence (CA8), and two individual confidence measures, LRD and MLM [51]. The sparsification curve for an ideal confidence map is described as optimal.

Fig. 8 presents the sparsification curves for enhanced confidence, confidence measure combinations, and high-performing individual confidence measures across multiple datasets. As the proportion of removed low-confidence points increases, the COMP values for C22, C8, and CA8 rise significantly.

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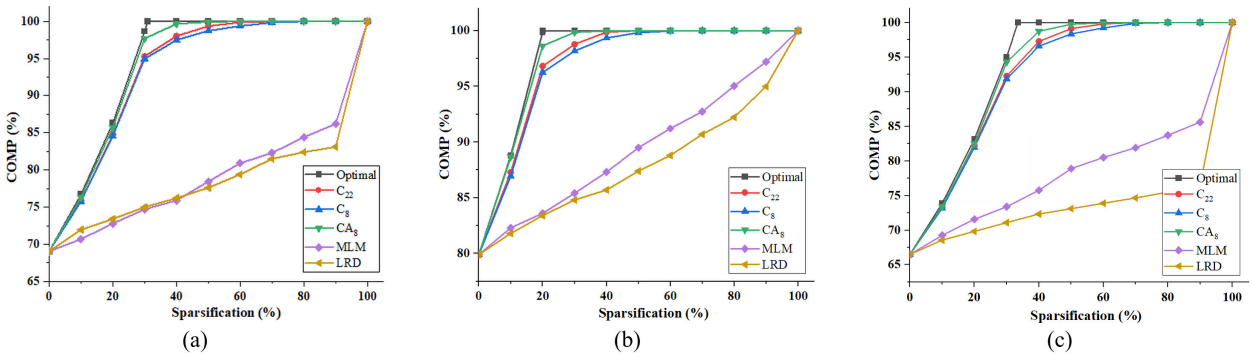


Fig. 8. Comparison of sparsification curves for multiple datasets. (a) US3D_JAX, (b) US3D_OMA, and (c) MVS3D.

TABLE V
EVALUATION RESULTS OF MULTIPLE DSM FUSION METHODS

Data/Method		Our	MF [55]	WA [19]	ADF [20]	IBF [17]	UGDF [24]
US3D_JAX (Dataset2)	MAE(m)	1.64	1.98	2.11	1.90	1.77	**1.88**
	RMSE(m)	3.15	4.02	4.38	3.84	3.46	**3.73**
	MHE(m)	0.58	0.91	1.00	0.85	0.71	**0.79**
	COMP(%)	57.81	49.73	46.69	51.24	54.52	**52.64**
US3D_OMA (Dataset3)	MAE(m)	0.83	1.06	1.13	1.01	0.89	**0.93**
	RMSE(m)	1.78	2.19	2.42	2.06	1.93	**1.98**
	MHE(m)	0.34	0.56	0.65	0.49	0.40	**0.45**
	COMP(%)	80.10	75.47	73.28	76.38	78.55	**77.62**
MVS3D (Dataset6)	MAE(m)	1.94	2.53	2.75	2.33	2.12	**2.24**
	RMSE(m)	3.67	4.94	5.36	4.48	3.93	**4.27**
	MHE(m)	0.56	1.17	1.40	0.92	0.70	**0.83**
	COMP(%)	62.79	51.26	45.97	55.43	60.29	**57.51**

The similarity between the C22 and C8 curves suggests that C8 effectively captures most of the information from C22 after confidence measure selection. CA8 outperforms C22, with its sparsification curve exhibiting a faster increase in COMP values. In contrast, the individual confidence measures, LRD and MLM, show instability and fail to provide reliable reconstruction accuracy predictions across diverse scenarios. Furthermore, Fig. 8 demonstrates that for the challenging MVS3D dataset, the enhanced confidence maintains a strong representational capability in reconstruction accuracy. These findings validate that the adopted enhanced confidence exhibits strong robustness and effectively guides DSM fusion.

F. Analysis of DSM Fusion Methods

To evaluate the performance of the proposed DSM fusion strategy, a quantitative comparison was conducted against mainstream DSM fusion methods, including MF [55], WA [19], ADF [20], iterative bilateral filtering (IBF) [17], and UGDF [24]. The experiments were performed on the US3D_JAX, US3D_OMA, and MVS3D datasets to assess the effectiveness of different DSM fusion methods.

Table V presents the fusion results of the proposed confidence-guided DSM fusion strategy compared with mainstream DSM fusion methods across multiple datasets. The red figures rank first, the blue figures rank second, and the bold figures rank third in each metric for each dataset. As shown in Table V, the proposed method achieves superior reconstruction performance, outperforming existing approaches in terms of accuracy, stability, and completeness. MF and WA exhibit the weakest performance. MF is highly susceptible to erroneous observations, while WA struggles to obtain accurate weighting information, leading to bias. Additionally, WA assigns weights based on the performance of individual DSMs, but since different regions of a DSM contribute unequally to reconstruction quality, this method may discard valuable information from low-weight DSMs. When a larger number of DSMs are available, the aggregated height values exhibit certain statistical properties, resulting in MF performing better than WA. ADF improves performance by incorporating spatial correlation, defining an irregular region around each pixel based on spatial and color proximity, and applying MF within this region, yielding results superior to MF. UGDF further refines MF by integrating matching uncertainty into the fusion process, leading to better performance than ADF. IBF applies the bilateral filtering framework in an iterative scheme and achieves the best performance among the mainstream DSM fusion methods. The proposed confidence-guided DSM fusion method effectively leverages the strengths of multiple DSMs, significantly enhancing the accuracy and completeness of 3-D reconstruction in urban areas.

G. Analysis of End-to-End Performance

To effectively assess the proposed satellite MVS method, mainstream comparative methods employed include: SatMVSF (REDNet) [16], SatMVSF (CasMVSNet) [16], SatMVSF (UCSNet) [16], Satellite Stereo Pipeline (S2P) [33],

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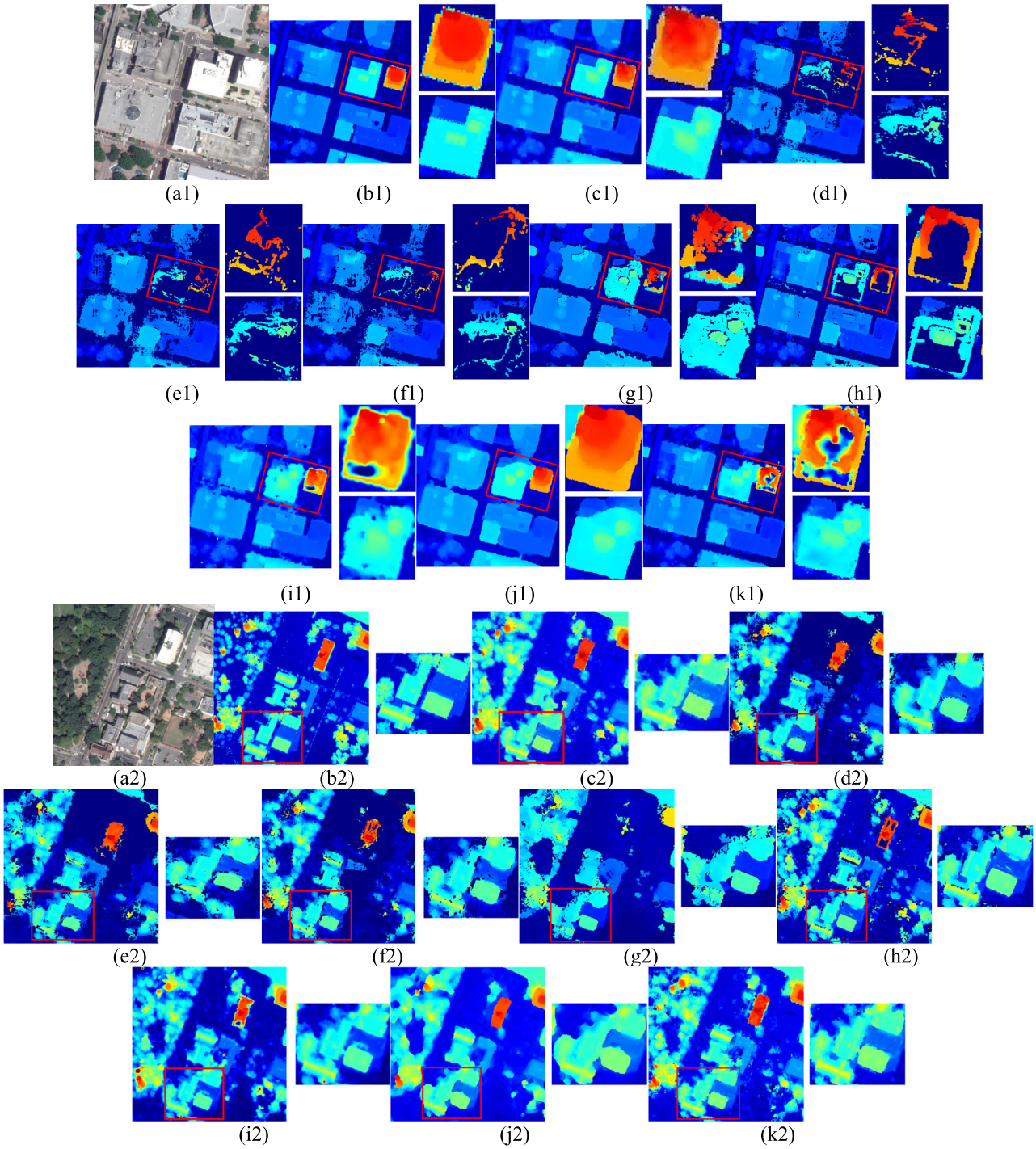


Fig. 9. Visualization results of the proposed method compared with the mainstream multiview 3-D reconstruction methods on the US3D_JAX dataset. Dataset1 and Dataset2 (a1) and (a2) satellite images, (b1) and (b2) GT, (c1) and (c2) proposed method, (d1) and (d2) SatMVSF (REDNet), (e1) and (e2) SatMVSF (CasMVSNet), (f1) and (f2) SatMVSF (UCSNet), (g1) and (g2) S2P, (h1) and (h2) Adapted COLMAP, (i1) and (i2) Catalyst, (j1) and (j2) Smart3D Sat, and (k1) and (k2) Agisoft.

Adapted COLMAP [18], Catalyst [39], Smart3D Sat (https://www.soarscape.com), and Agisoft [10].

SatMVSF (REDNet), SatMVSF (CasMVSNet), and SatMVSF (UCSNet) are three widely used learning-based methods for MVS. The SatMVSF framework, proposed by Gao et al. [16], was designed for 3-D reconstruction of the Earth's surface from multiview optical satellite imagery. Because Gao et al. [16] noted the limited size of the MVS3D dataset and the scene variations in the US3D dataset, these two datasets were not used for training and were only used to assess generalization. In accordance with that practice, the trained models with default parameters were used for prediction in this study.

S2P is an automatic and modular stereo pipeline designed for push-broom images. To improve computational efficiency, the input remote sensing images were tiled into smaller blocks

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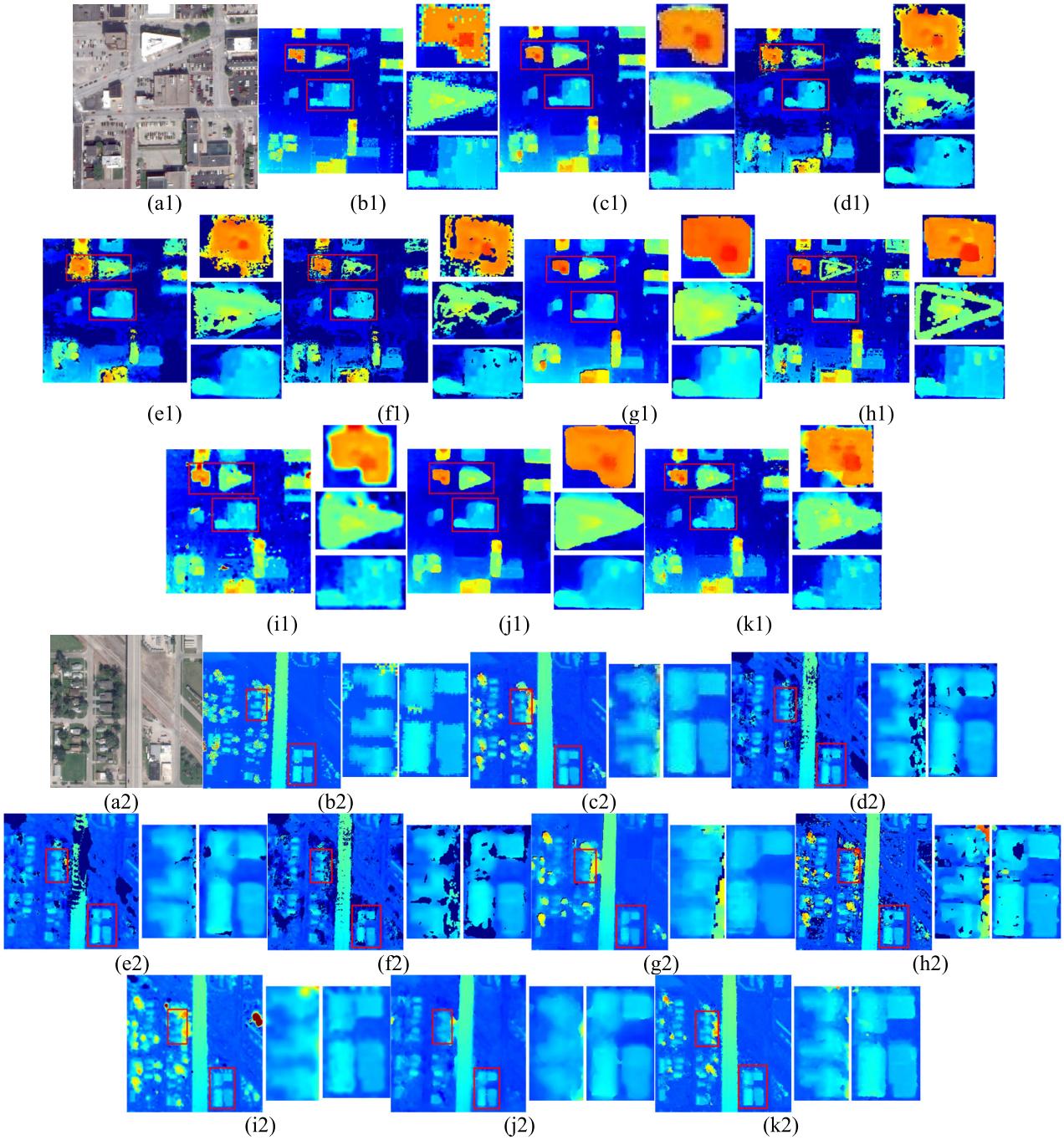


Fig. 10. Visualization results of the proposed method compared with the mainstream multiview 3-D reconstruction methods on the US3D_OMA dataset. Dataset3 and Dataset4 (a1) and (a2) satellite images, (b1) and (b2) GT, (c1) and (c2) proposed method, (d1) and (d2) SatMVSF (REDNet), (e1) and (e2) SatMVSF (CasMVSNet), (f1) and (f2) SatMVSF (UCSNet), (g1) and (g2) S2P, (h1) and (h2) Adapted COLMAP, (i1) and (i2) Catalyst, (j1) and (j2) Smart3D Sat, and (k1) and (k2) Agisoft.

for parallel processing. In our experiments, the outlier removal threshold in the fusion stage was set to 3.1 m (about ten times the image resolution as suggested by the original S2P). The default dense matching algorithm in S2P is MGM [34], and all other parameters were kept at their defaults [56].

Adapted COLMAP utilizes the reconstruction pipeline from the computer vision community for 3-D reconstruction, where the RPC model of the local satellite image patches is fit by a pinhole camera model. Because Adapted COLMAP does not support large scenes, the imagery was cropped into tiles to produce a series of small DSM blocks, which were mosaicked into the final DSM. Default parameters were used.

Catalyst is a professional remote sensing and photogrammetry suite developed by PCI Geomatics for geometric processing and 3-D modeling of high-resolution satellite imagery. In our experiments, the SGM algorithm was selected, and all other parameters were kept at their default values.

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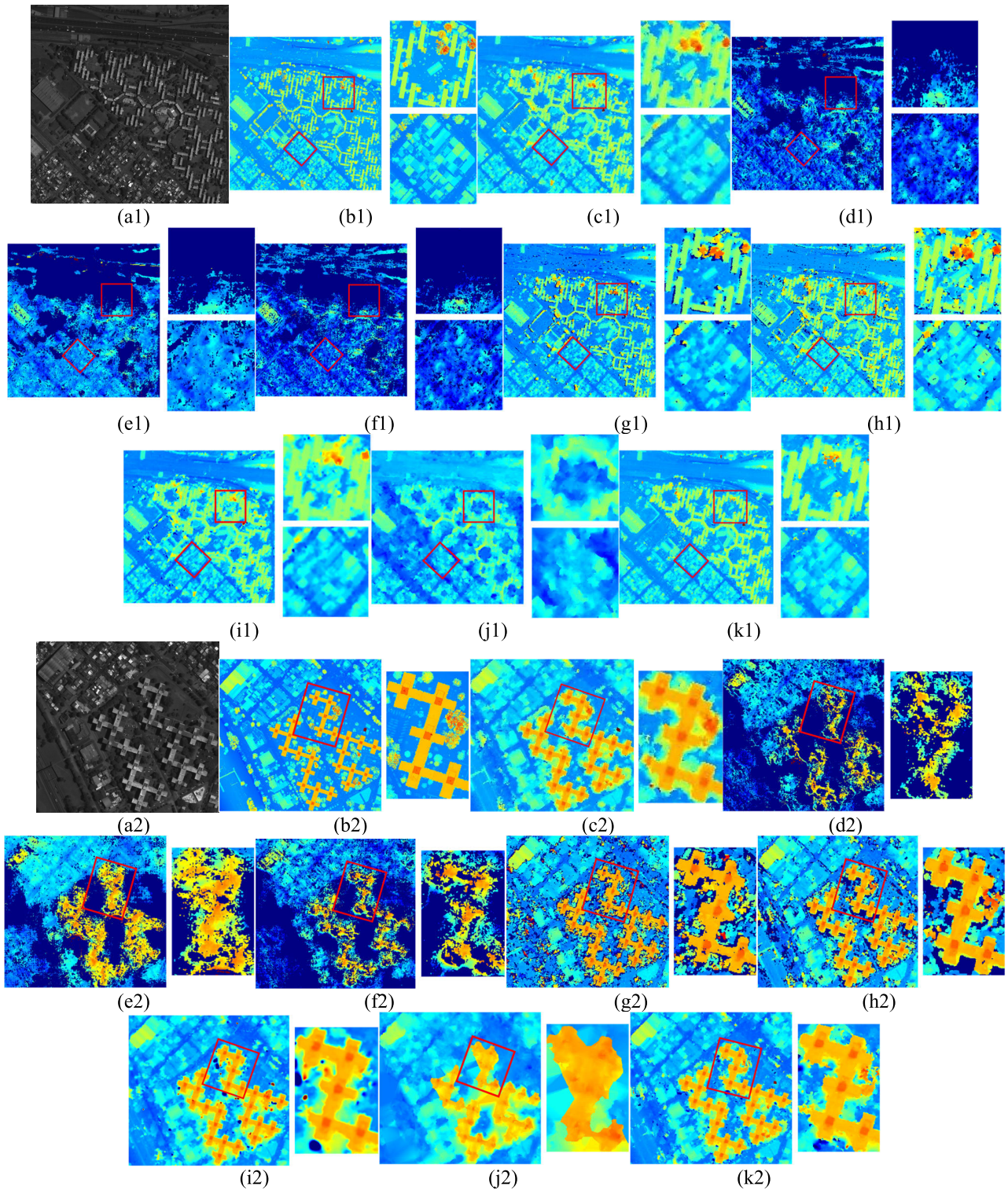


Fig. 11. Visualization results of the proposed method compared with the mainstream multiview 3-D reconstruction methods on the MVS3D dataset. Dataset5 and Dataset6 (a1) and (a2) satellite images, (b1) and (b2) GT, (c1) and (c2) proposed method, (d1) and (d2) SatMVSF (REDNet), (e1) and (e2) SatMVSF (CasMVSNet), (f1) and (f2) SatMVSF (UCSNet), (g1) and (g2) S2P, (h1) and (h2) Adapted COLMAP, (i1) and (i2) Catalyst, (j1) and (j2) Smart3D Sat, and (k1) and (k2) Agisoft.

Smart3D Sat is a satellite imagery-based 3-D modeling software developed by SoarScape, China. It includes a proprietary dense matching algorithm with high robustness and supports one-click automated model generation. Default parameters were used.

Agisoft supports multiview satellite image processing. To improve reconstruction quality, the ultrahigh accuracy option was selected for photograph alignment and dense matching, and the remaining parameters were kept at their defaults.

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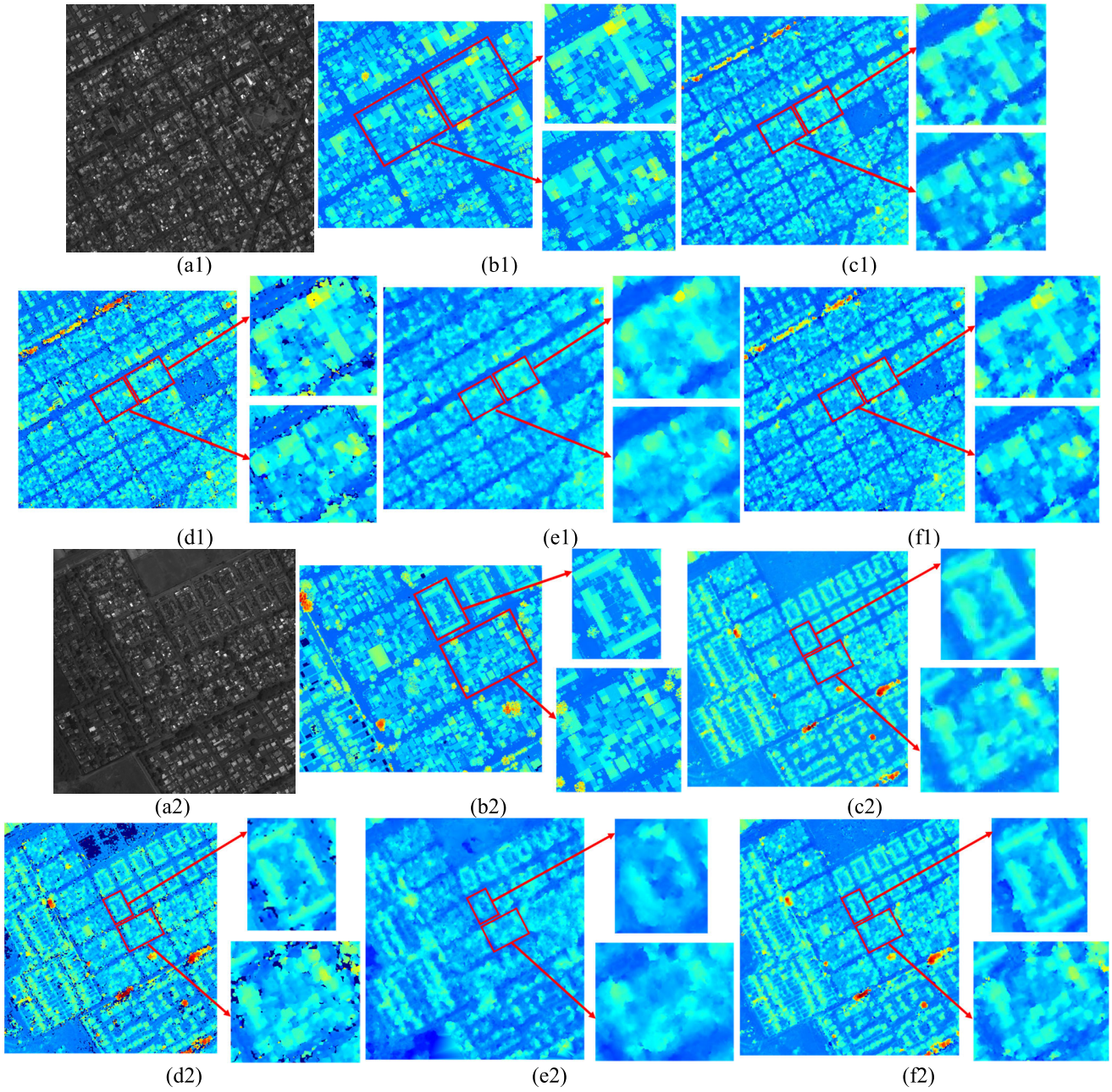


Fig. 12. Visualization results of the proposed method compared with representative multiview 3-D reconstruction methods in large-scale urban areas. Region1 and Region2 (a1) and (a2) satellite images, (b1) and (b2) GT, (c1) and (c2) proposed method, (d1) and (d2) Adapted COLMAP, (e1) and (e2) Smart3D Sat, and (f1) and (f2) Agisoft.

Figs. 9–11 visualize the reconstruction results of the proposed method compared to mainstream multiview 3-D reconstruction methods. Dataset1 and Dataset2, derived from the US3D_JAX dataset, include high-rise buildings, complex structures, and texture-less buildings. The proposed method effectively integrates complementary information from multiview images, yielding more complete and accurate reconstructions with well-preserved building details and clearer edge contours, closely aligning with the GT. In contrast, the three deep learning-based frameworks exhibit inferior performance, with substantial void regions and poor reconstruction quality, particularly for texture-less and high-rise structures. SatMVSF (REDNet) performs slightly better than the other two deep learning models. Among traditional methods, S2P struggles with texture-less targets, often missing structural details and producing blurred edges. Adapted COLMAP performs poorly in texture-less regions but better preserves building details and contours. Among mainstream software methods, Smart3D Sat delivers the most complete building reconstructions with smooth surfaces. However, it tends to exhibit issues such as building merging and blurred details. Catalyst exhibits relatively poor completeness in reconstructing building structures and generates extensive noisy regions. Agisoft introduces slight noise in

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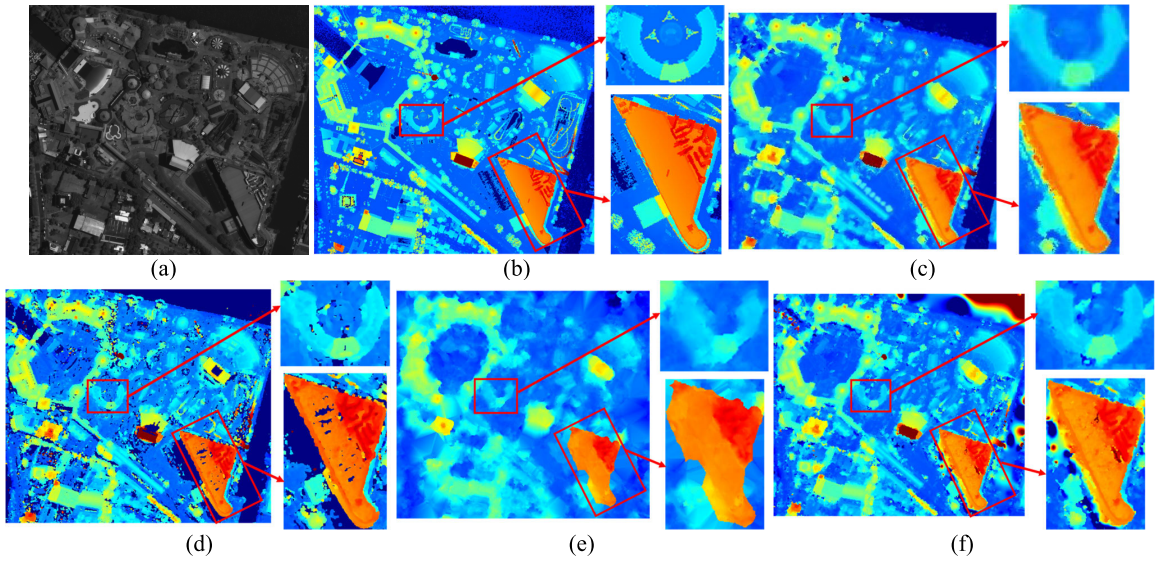


Fig. 13. Visualization results of the proposed method compared with representative multiview 3-D reconstruction methods in large-scale urban areas. Region3 (a) satellite images, (b) GT, (c) proposed method, (d) Adapted COLMAP, (e) Smart3D Sat, and (f) Agisoft.

TABLE VI

EVALUATION RESULTS OF THE PROPOSED METHOD COMPARED WITH MAINSTREAM METHODS ON DIFFERENT DATASETS

Method/ Dataset		Our	SatMVSF (REDNet)	SatMVSF (CasMVSNet)	SatMVSF (UCSNet)	S2P	Adapted COLMAP	Catalyst	Smart3D Sat	Agisoft
1	MAE(m)	1.20	2.67	3.28	2.43	1.84	**1.64**	1.86	1.48	1.79
	RMSE(m)	2.02	5.99	6.72	5.03	3.92	**3.14**	3.65	2.73	3.38
	MHE(m)	0.40	0.67	0.69	0.77	0.39	**0.45**	0.57	0.46	0.53
	COMP(%)	69.07	35.95	30.01	30.96	58.56	50.84	56.29	64.62	**61.22**
2	MAE(m)	1.64	2.66	3.01	2.99	3.86	**2.41**	2.75	1.83	2.68
	RMSE(m)	3.15	4.28	4.75	4.81	6.23	**4.17**	4.54	3.51	4.37
	MHE(m)	0.58	1.14	1.31	1.31	1.36	**0.72**	1.18	0.66	1.14
	COMP(%)	57.81	27.87	22.32	23.73	19.47	46.44	43.60	52.79	**46.88**
3	MAE(m)	0.83	1.56	2.12	1.84	**1.00**	1.18	1.57	0.81	1.44
	RMSE(m)	1.78	3.43	4.56	3.78	**2.20**	2.91	3.43	1.74	3.02
	MHE(m)	**0.34**	0.54	0.59	0.60	0.39	0.28	0.49	0.26	0.44
	COMP(%)	80.10	40.29	38.74	32.32	**75.76**	63.78	67.81	83.12	72.29
4	MAE(m)	0.72	1.04	1.13	1.16	1.15	1.21	1.18	**1.07**	1.04
	RMSE(m)	1.59	2.20	2.42	2.36	**2.32**	2.67	2.59	2.53	2.44
	MHE(m)	0.20	0.48	0.45	0.53	0.49	0.33	0.40	**0.28**	0.24
	COMP(%)	85.53	63.77	64.76	57.35	71.02	61.05	67.45	**75.37**	78.41
5	MAE(m)	1.53	4.33	3.55	4.73	2.02	1.69	1.94	2.48	**1.86**
	RMSE(m)	2.82	5.71	4.93	6.22	3.44	3.12	3.52	4.33	**3.38**
	MHE(m)	0.47	3.34	2.39	3.75	0.81	0.53	0.59	1.16	**0.55**
	COMP(%)	66.34	9.77	12.45	9.18	53.16	59.39	**59.86**	42.49	61.35
6	MAE(m)	1.94	4.53	3.58	4.49	2.77	2.16	2.45	2.72	**2.34**
	RMSE(m)	3.67	6.50	5.11	6.15	4.73	4.28	4.51	4.67	**4.36**
	MHE(m)	0.56	2.88	2.41	3.12	1.10	0.62	0.78	0.94	**0.71**
	COMP(%)	62.79	7.30	9.45	6.71	37.67	40.27	**53.40**	39.18	56.71
Mean	MAE(m)	1.31	2.80	2.78	2.94	2.11	1.72	1.96	**1.73**	1.86
	RMSE(m)	2.51	4.69	4.75	4.73	3.81	**3.38**	3.71	3.25	3.49
	MHE(m)	0.43	1.51	1.31	1.68	0.76	0.49	0.67	0.63	**0.60**
	COMP(%)	70.27	30.83	29.62	26.71	52.61	53.63	58.07	**59.60**	62.81

TABLE VII

EVALUATION RESULTS OF THE PROPOSED METHOD COMPARED WITH MAINSTREAM METHODS IN LARGE-SCALE URBAN AREAS

Method		Our	Adapted COLMAP	Smart 3D Sat	Agisoft
Region1	MAE(m)	1.38	1.48	1.87	**1.51**
	RMSE(m)	2.49	2.69	3.46	**2.78**
	MHE(m)	0.42	0.45	0.63	**0.47**
	COMP(%)	75.51	**67.84**	54.88	70.19
Region2	MAE(m)	1.46	**1.59**	2.13	1.54
	RMSE(m)	2.62	**2.93**	3.87	2.85
	MHE(m)	0.47	**0.56**	0.82	0.52
	COMP(%)	71.84	**61.38**	44.90	66.86
Region3	MAE(m)	2.27	2.38	3.14	**2.52**
	RMSE(m)	3.96	4.25	5.35	**4.49**
	MHE(m)	0.70	0.72	1.19	**0.78**
	COMP(%)	57.98	51.07	35.64	**47.96**

certain areas and shows reduced accuracy in texture-less regions.

Dataset3 and Dataset4, derived from the US3D_OMA dataset, contain texture-less buildings, complex structures, and small targets. The proposed method consistently achieves superior reconstruction results across various urban scenarios, effectively preserving the structural integrity, details, and contours of building targets. However, the deep learning-based methods generate large void regions, exhibit poor building completeness, and suffer from edge blurring. Among traditional methods, S2P performs well, introducing minimal noise and maintaining high completeness in urban reconstructions, though some details are still missing. Adapted COLMAP exhibits numerous void regions and struggles with texture-less buildings and urban vegetation. Among mainstream software methods, Smart3D Sat produces reconstructions with well-defined building outlines and smooth surfaces, though it often

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TABLE VIII
COMPARISON OF THE TIME CONSUMPTION OF THE PROPOSED METHOD AND THE MAINSTREAM METHODS

Method	Our	SatMVSF (REDNet)	SatMVSF (CasMVSNet)	SatMVSF (UCSNet)	S2P	Adapted COLMAP	Catalyst	Smart3D Sat	Agisoft
Times	7min53s	4min17s	3min32s	3min44s	1min24s	4min46s	46min25s	54min37s	1min58s

fails to preserve fine object details. Catalyst tends to generate numerous noisy regions in its reconstructions. Agisoft delivers relatively stable performance.

Dataset5 and Dataset6, derived from the MVS3D dataset, contain complex building clusters and small structures, presenting a highly challenging urban scenario. The proposed method produces reconstructions that closely align with the GT, achieving high completeness for building clusters while effectively preserving details and accurately reconstructing small structures. In contrast, the three deep learning-based methods fail to reconstruct complex building clusters effectively, highlighting their limited generalization capability. Among traditional methods, S2P produces relatively complete building reconstructions but introduces substantial noise and void regions. Adapted COLMAP exhibits numerous void regions but performs well in reconstructing complex building clusters. Among mainstream software methods, Agisoft achieves high completeness in reconstructing complex building clusters but introduces slight noise and loses some structural details. Catalyst demonstrates good performance in reconstructing building details, although its overall results are slightly inferior to those of Agisoft. Smart3D Sat performs poorly in complex urban areas, exhibiting blurred building structures, missing object details, and excessive surface smoothing.

A comprehensive quantitative evaluation was conducted using MAE, RMSE, MHE, and COMP to compare the proposed satellite MVS method with the mainstream methods. MAE and RMSE measure reconstruction accuracy, MHE reflects stability, and COMP assesses completeness. Table VI presents the quantitative evaluation results for the proposed method and mainstream methods across various datasets. The red figures rank first, the blue figures rank second, and the bold figures rank third in each metric for each dataset. Across the US3D and MVS3D datasets, the proposed method outperforms mainstream methods in terms of accuracy, stability, and completeness, achieving an average MAE of 1.31 m, RMSE of 2.51 m, MHE of 0.43 m, and COMP of 70.27%. The deep learning methods exhibit poor stability and completeness. Specifically, SatMVSF (REDNet) achieves an average MAE of 2.80 m, RMSE of 4.69 m, MHE of 1.51 m, and COMP of 30.83%. SatMVSF (CasMVSNet) attains an average MAE of 2.78 m, RMSE of 4.75 m, MHE of 1.31 m, and COMP of 29.62%. SatMVSF (UCSNet) yields an average MAE of 2.94 m, RMSE of 4.73 m, MHE of 1.68 m, and COMP of 26.71%. Among traditional methods, Adapted COLMAP demonstrates the best performance, achieving an average MAE of 1.72 m, RMSE of 3.38 m, MHE of 0.49 m, and COMP of 53.63%. S2P attains an average MAE of 2.11 m, RMSE of 3.81 m, MHE of 0.76 m, and COMP of 52.61%. Among mainstream software methods, Catalyst achieves an average MAE of 1.96 m, RMSE of 3.71 m, MHE of 0.67 m, and COMP of 58.07%. Smart3D Sat attains an average MAE of 1.73 m, RMSE of 3.25 m, MHE of 0.63 m, and COMP of 59.60%. Agisoft yields an average MAE of 1.86 m, RMSE of 3.49 m, MHE of 0.60 m, and COMP of 62.81%. Overall, the proposed method demonstrates stable and accurate reconstruction results across various urban scenarios, exhibiting strong robustness. Additionally, Adapted COLMAP, Smart3D Sat, and Agisoft also achieve competitive performance across multiple datasets.

To evaluate the 3-D reconstruction performance of the proposed method in large-scale urban scenes, three urban regions (Region1–Region3) were selected for comparative analysis against several state-of-the-art methods. Figs. 12 and 13 visualize the reconstruction results of the proposed method alongside those of Adapted COLMAP, Smart3D Sat, and Agisoft across Region1–Region3. Table VII provides a quantitative comparison of reconstruction performance in these large-scale urban areas. The red figures rank first, the blue figures rank second, and the bold figures rank third in each metric for each dataset. The proposed method demonstrates superior reconstruction accuracy and completeness in large-scale urban scenes, yielding results that are closer to the GT and better preserve building details and structural outlines. Among the compared methods, Adapted COLMAP provides detailed reconstruction of buildings and urban elements but suffers from limited completeness. Agisoft effectively reconstructs 3-D urban models, though its results exhibit a certain degree of noise. Smart3D Sat performs poorly in dense or complex building regions, often producing oversmoothed outputs with missing details of buildings and ground objects, leading to low reconstruction accuracy and completeness. In summary, the proposed method exhibits strong robustness and adaptability in large-scale and diverse urban scenes.

The average computational time required to process multiview satellite images covering regions of 2000×2000 to 3000×3000 pixels is presented in Table VIII. Among traditional methods, S2P has an average processing time of 1 min 24 s, while Adapted COLMAP takes 4 min 46 s. Among mainstream software methods, Catalyst requires an average processing time of 46 min 25 s, Smart3D Sat requires 54 min 37 s, and Agisoft requires 1 min 58 s. The three deep learning methods, SatMVSF (REDNet), SatMVSF (CasMVSNet), and SatMVSF (UCSNet), which utilize pretrained models, require 4 min 17 s, 3 min 32 s, and 3 min 44 s, respectively. The proposed method achieves an average processing time of 7 min 53 s, which remains within an acceptable range.

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V. Conclusion and Future Works

We propose a satellite MVS method that improves urban 3-D reconstruction by combining optimized image combination selection with confidence-guided DSM fusion. To reduce failures caused by redundant stereo pairs or insufficient spatial coverage, an image selection approach driven by satellite image meta-parameters is introduced that integrates rule-based partitioning with a hierarchical heuristic strategy. To address the limitations of conventional DSM fusion, which often fails to distinguish DSM quality and is susceptible to occlusions and noise, a confidence-guided fusion scheme is designed that balances local statistical characteristics with single-point confidence. Experiments across various urban scenes show consistent improvements, achieving an average MAE of 1.31 m, RMSE of 2.51 m, MHE of 0.43 m, and COMP of 70.27%.

To meet the demands of realistic 3-D urban reconstruction, future research will focus on further improving reconstruction quality across diverse urban environments. In addition, future work will explore 3-D reconstruction of building targets using primitive model constraints and texture mapping.

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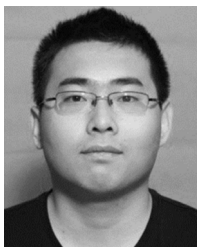
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